

simplePHENOTYPES — Equation-to-Code Audit Map

Every theory-derived equation, its primary reference, and where it is implemented (function, file, line)

Independent dual-model audit (Claude Fable + OpenAI Codex), assembled from per-group reports

2026-09-29 audit at c511c6f; V2 line references re-checked against 20cd2c9 (2026-10-08);
V1 references as remapped to e666a2c (2026-10-04)

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How to read this document

Each entry gives the **equation as implemented**, the **variables**, the **primary reference** (a page or equation number is quoted only when the auditor verified it; otherwise it is marked *page unverified*), the **implementing function**, and the **exact file:line range**. V1 ranges are at e666a2c (remapped by git line mapping from the audit commit c511c6f; see Appendix C). V2 ranges, and the V2 rows of Appendix B, are at 20cd2c9 (2026-10-08): every V2

Location was re-mapped from the commit it was actually written against (the audit commit for :NNN shorthands, e666a2c or 4959077 for file-qualified ranges) and checked against the code at 20cd2c9 (content of the mapped lines, plus an automated check that the first R range of each entry falls inside its named function). Line hints inside *Notes* fields were updated only where an entry was revised (selection, OCS/usefulness, crossing, transcriptome); other *Notes* hints may still refer to c511c6f. Appendix A is the original automated check at the audit commit.

Revision 2026-10-08 (equations changed in code after the audit). The following entries were rewritten to match the current source; their page-verification status is unchanged: V2-crossing-3 (default meiosis is the two-pathway gamma interference model, $\nu = 2.6$, $p = 0$, since 2.0.0.9003, DECISION-047; Poisson is interference = "poisson"), with Poisson-only notes on V2-crossing-4/5/9 and V2-rust-core-1/13; V2-transcriptome-12 (realized cis fraction $v_{cis}/\text{Var}(G)$); V2-transcriptome-19 ($\hat{\kappa}$ now subtracts the GREML-implied genetic trans share); V2-transcriptome-20 (observe_counts carries @references); V2-transcriptome-C2 (TX-F2 note); V2-grammar-25 and V2-transcriptome-25 (polynomial hash of the layer type in the sub-seed); V2-grammar-10 and V2-effects-arch-18 (residual_mode); V2-effects-arch-20 (partner = "random"); V2-selection-15 (multinomial bulk); V2-ocs-usefulness-marker-8 (sample_parents() allocates by default).

Conventions: dosage is coded -1/0/1 unless stated; p is the allele frequency of the +1 allele; q = 1 - p. "Own design" means the equation is a package design decision (documented in docs/DECISIONS.md) rather than a literature formula.

Sections V2 and V1 are separate: V1 is the frozen legacy create_phenotypes() engine (DECISION-003/008), V2 is the composable grammar, crossing, selection, prediction and transcriptome layers.

Version 2 (grammar, crossing, selection, prediction, transcriptome, I/O)

Grammar: variance budget, layers, realization, orthogonal model, vQTL, seeds

Source report: v2-grammar.md

Entries: 28

V2-grammar-1. Layer scaling to prop

$$\tilde{c} = \frac{c - \bar{c}}{\text{sd}(c)} \sqrt{\pi_\ell}$$

- *Variables:* c raw component, π_ℓ layer prop
- *Reference:* Package convention (SPEC §2)
- *Function:* .genetic_matrix
- *Location:* R/grammar_realize.R:139-161 (146)
- *Notes:* Sample sd (n-1). Layers summed; no orthogonalization.

V2-grammar-2. Additive component

$$c_A = \sum_j e_j x_{ij}, x \in \{-1, 0, 1\}$$

- *Variables:* e geometric/custom effects
- *Reference:* Fernandes & Lipka 2020 (v1 convention)
- *Function:* .component_raw
- *Location:* R/grammar_realize.R:498-509 (500), centred 432

V2-grammar-3. Orthogonal genotypic value

$$g_{ij} = a_j x_{ij} + d_j \mathbb{1}[x_{ij} = 0] \text{ i.e. } -a_j/ + d_j/ + a_j$$

- *Variables:* a, d per locus
- *Reference:* Falconer & Mackay 1996 (genotypic values; page unverified)
- *Function:* `.component_raw`
- *Location:* R/grammar_realize.R:501-507 (506)
- *Notes:* DECISION-020

V2-grammar-4. Dominance component

$$c_D = \sum_j e_j \mathbb{1}[x_{ij} = 0]$$

- *Variables:* het indicator
- *Reference:* Package convention
- *Function:* `.component_raw`
- *Location:* R/grammar_realize.R:510
- *Notes:* Not Fisher's D; see GRAM-F1

V2-grammar-5. Epistasis unit column

$$z_i = \prod_k (t_{ik} - \bar{t}_k), t_k = \text{dosage ("a")} \text{ or het indicator ("d")}$$

- *Variables:* interaction positions k
- *Reference:* Package convention (centred products; cf. Cockerham/NOIA only under HWE+LE)
- *Function:* `.epi_unit_column`
- *Location:* R/grammar_realize.R:579-589 (586, 588)
- *Notes:* Single source for realization and pleio normalizer

V2-grammar-6. Epistasis component

$$c_E = \sum_p e_p z_{ip}$$

- *Variables:* e per set
- *Reference:* as above
- *Function:* `.component_raw`
- *Location:* R/grammar_realize.R:511-529 (524)

V2-grammar-7. Geometric effect series

$$e_k = b^k, k = 1..n, \text{ default } b = 0.5$$

- *Variables:* b base
- *Reference:* Fernandes & Lipka 2020 (v1 geometric)
- *Function:* `.effect_series`
- *Location:* R/effects_series.R:23-72 (64)
- *Notes:* length-n vector used verbatim

V2-grammar-8. Repulsion phase

$$e_k \leftarrow e_k (-1)^{k+1}$$

- *Reference:* Package convention
- *Function:* `.apply_phase`
- *Location:* R/grammar_layers.R:1482-1490 (1487)
- *Notes:* positional, not LD-derived (documented)

V2-grammar-9. Residual budget

$$\sigma_e^2 = \max(0, 1 - \sum_{\ell} \pi_{\ell} - \sum_v \pi_v)$$

- *Variables:* mean-effect and vqtl props
- *Reference:* SPEC §2
- *Function:* `.realize_phenotype`
- *Location:* R/grammar_realize.R:55-59 (59)

V2-grammar-10. Residual draw

$$e \sim N(0, \sigma_e^2) \text{ then } e \leftarrow \frac{e - \bar{e}}{\text{sd}(e)} \sigma_e$$

- *Reference:* Package convention
- *Function:* `.draw_residual`
- *Location:* R/effects_series.R:85-102 (99)
- *Notes:* exact sample variance; same standardized vector per (seed,trait,rep). Default `residual_mode = "fixed"`; `residual_mode = "random"` (added after the audit) returns the raw $N(0, \sigma_e^2)$ draw (:94)

V2-grammar-11. Phenotype

$$y_{it} = G_{it} + T_{it} + e_{it} + \mu_t$$

- *Variables:* G marker genetic, T transcriptome, μ mean
- *Reference:* —
- *Function:* `.realize_phenotype`
- *Location:* R/grammar_realize.R:96

V2-grammar-12. Requested budget identity

$$\sum_{A,D,E} \pi = h^2; \sum_{\text{all}} \pi \leq 1$$

- *Reference:* SPEC §2
- *Function:* `.resolve_prop`, `.add_layer`, `.check_h2_complete`
- *Location:* R/grammar_layers.R:1085-1098, 1102-1115; R/grammar_realize.R:1153-1166
- *Notes:* tol 1e-8

V2-grammar-13. One-call split

$$\pi_c = h^2/|\text{model}|$$

- *Variables*: components A,D,E
- *Reference*: SPEC §4.1
- *Function*: `.build_one_call`
- *Location*: R/grammar_simulate_phenotype.R:767-780 (769)
- *Notes*: “AE”: `n_pairs = n_qtn`

V2-grammar-14. Realized H²

$$\hat{H}_t^2 = \frac{1}{R} \sum_r \frac{\text{Var}(G_{.tr})}{\text{Var}(y_{.tr})}$$

- *Variables*: G = marker + genetic-mediated Tx
- *Reference*: SPEC §2 (V2)
- *Function*: `.realized_h2`
- *Location*: R/grammar_realize.R:1078-1106 (716-720)
- *Notes*: `vqtl` in denominator only

V2-grammar-15. Average effect of substitution

$$\alpha_j = a_j + d_j(q_j - p_j) = a_j + d_j(1 - 2p_j)$$

- *Variables*: p freq of +1 allele
- *Reference*: Falconer 1985 *Genet. Res.* 46:337-347; Falconer & Mackay 1996 (page unverified); Fisher 1918
- *Function*: `.avg_effect`
- *Location*: R/grammar_realize.R:182-184 (183)
- *Notes*: verified numerically

V2-grammar-16. Breeding value

$$A_i = \sum_j \alpha_j(x_{ij} - 2p_j), x \in \{0, 1, 2\}$$

- *Reference*: Falconer & Mackay 1996; Lynch & Walsh 1998
- *Function*: `.breeding_value_matrix`
- *Location*: R/grammar_realize.R:282-303 (227-235)
- *Notes*: a, d rescaled by $\sqrt{\pi}/\text{sd}$ (273); DECISION-019

V2-grammar-17. Orthogonal split

$$\text{Var}(g) = \text{Var}(A) + \text{Var}(D) + 2\text{Cov}(A, D); \text{ rows } \frac{\text{Var}(A)}{\text{Var}(g)}, \frac{\text{Var}(D)}{\text{Var}(g)}, 1 - \text{add} - \text{dom}$$

- *Variables*: D = g – A
- *Reference*: Fisher 1918; Falconer & Mackay 1996

- *Function:* `.orthogonal_var_split`, `.variance_budget`
- *Location:* R/grammar_realize.R:778-794 (646-650), 536-551
- *Notes:* Cov = 0 under random mating (verified)

V2-grammar-18. HWE theory (check only)

$$V_A = \sum 2p_j q_j \alpha_j^2, V_D = \sum (2p_j q_j d_j)^2$$

- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* — (audit check)
- *Location:* —
- *Notes:* matches 0.8536 vs 0.8539

V2-grammar-19. vQTL loading

$$L_i = \sum_v \sqrt{\pi_v} \frac{s_{iv} - \bar{s}_v}{\text{sd}(s_v)}, s_v = X_v e_v$$

- *Reference:* Package choice (DGLM-style log link; Rönnegård & Valdar 2011, Murphy et al. 2022 cited for context only)
- *Function:* `.apply_vqtl`
- *Location:* R/grammar_realize.R:605-618 (616)

V2-grammar-20. vQTL residual

$$\log \text{Var}(E_v | g) = \text{const} + L_i; E_{vi} = z_i e^{L_i/2}, \text{ then centred and scaled to } \sqrt{\pi_v}$$

- *Variables:* z standardized normal
- *Reference:* as above
- *Function:* `.apply_vqtl`
- *Location:* R/grammar_realize.R:627-641
- *Notes:* verified numerically

V2-grammar-21. Transcriptome score

$$c_T = \sum_g w_g z_g, z_g = (E_g - \bar{E}_g) / \text{sd}(E_g), w \leftarrow w / \max |w|; \text{ genetic part same denominator}$$

- *Reference:* SPEC-transcriptome (not this group)
- *Function:* `.tx_raw`, `.transcriptome_matrix`
- *Location:* R/grammar_realize.R:454-476 (470, 473, 475), 308-338

V2-grammar-22. Mediation split

$$\frac{\text{Var}(T_g)}{V_P}, \frac{\text{Var}(T_e)}{V_P}, \frac{2\text{Cov}(T_g, T_e)}{V_P}$$

- *Reference:* —
- *Function:* `.mediation_budget`

- *Location:* R/grammar_realize.R:738-750 (604-606)
- *Notes:* stale in complex (F3)

V2-grammar-23. Complex combination

$$G_t^{(c)} = \frac{\sum_m G_t^{(m)} - \bar{G}}{\text{sd}} \sqrt{h_t^2}; e \sim \text{residual with } 1 - h_t^2$$

- *Reference:* SPEC §4.3
- *Function:* complex_phenotypes
- *Location:* R/grammar_complex.R:122-134 (123, 127), 96-107 (99)
- *Notes:* inputs weighted by sd ($\propto \sqrt{\text{prop}}$)

V2-grammar-24. MAF

$$p_j = (x_j + 1)/2, \text{MAF} = \min(p, 1 - p)$$

- *Reference:* —
- *Function:* .marker_maf_ref (via .marker_stats_ref)
- *Location:* R/grammar_simulate_phenotype.R:1022-1052 (1040, 1044)
- *Notes:* candidate = MAF>0 (F2)

V2-grammar-25. Sub-seed

$$b \leftarrow (257b + u_k) \bmod 2147483629 \text{ over the UTF-8 codes } u_k \text{ of the layer type } (b_0 = 0); s_\ell = (1009s + 7919b + 104729 \text{occ}) \bmod (2^{31} - 1)$$

- *Reference:* Package scheme (SPEC §6)
- *Function:* .layer_seed
- *Location:* R/grammar_simulate_phenotype.R:1079-1094 (1083, 1092)
- *Notes:* no collisions found (audit, with the former $\sum \text{utf8}$ type term); the type term is now an order-sensitive polynomial hash (:1083–1086)

V2-grammar-26. Per-QTN variance

$$v_j = \frac{\pi_\ell}{\text{sd}(c)^2} \frac{\text{Var}(\text{col}_j)}{V_P^{\text{real}}}$$

- *Reference:* Package convention
- *Function:* .qtn_var
- *Location:* R/io_write.R:1288-1312 (217-218)
- *Notes:* marginal; sums $\neq$ prop under LD/shared loci

V2-grammar-27. Fixed-scale phenotype

$$y = g + e, e \sim N(0, \sigma_e^2), \sigma_e^2 = \text{Var}(g_{ref}) \frac{1 - h^2}{h^2}$$

- *Reference*: Falconer & Mackay 1996 ($h^2 = V_A/(V_A + V_E)$)
- *Function*: phenotype_value
- *Location*: R/cross_population.R:1004-1104 (1061)
- *Notes*: true normal, not standardized (DECISION-021)

V2-grammar-28. Genetic correlation (plot)

$$r = \text{cor}(G_{\cdot 1}, G_{\cdot 2})$$

- *Reference*: —
- *Function*: .plot_cor
- *Location*: R/grammar_plot.R:144-162 (150)
- *Notes*: rep 1 only

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-grammar-C1. PleioArch shared covariance

$$\Sigma_{ii} = \pi_i V_i, \quad \Sigma_{ij} = r_{ij} \sqrt{V_i V_j}$$

- *Variables*: π_i shared (pleiotropic) fraction, V_i layer variance (= prop), r_{ij} target cor
- *Reference*: Prado et al., in preparation (PleioArch); page unverified; reference code absent from checkout
- *Function*: .pleio_draw, .pleio_nonadditive_draw
- *Location*: R/effects_pleioarch.R:65-93 (sigma at 78-80), 345-358 (356-357)
- *Notes*: UNVERIFIABLE against primary source; internal form explicit. DECISION-007/013

V2-grammar-C2. PleioArch effect allocation (non-additive)

$$w_u \sim \text{MVN}(0, \Sigma/m), \quad b_u = w_u / \text{sd}(z_u) \text{ for the } m \text{ informative shared units; trait-specific units } \sim N(0, (1 - \pi_t)V_t)$$

- *Variables*: z_u realized unit design column, m shared units with $\text{sd} > 0$
- *Reference*: Prado et al., in preparation (additive form); non-additive extension is package design; page unverified
- *Function*: .pleio_unit_effects, .draw_mvnorm
- *Location*: R/effects_pleioarch.R:583-690 (535-537), 780-800
- *Notes*: Constant (hetless) units get effect 0 and are excluded from the allocation count

V2-grammar-C3. PleioArch attainability

$$M \succeq 0, \quad M_{ii} = \pi_i, \quad M_{ij} = r_{ij}; \text{ two traits: } r^2 \leq \pi_1 \pi_2$$

- *Variables*: variance-free feasibility matrix M
- *Reference*: Prado et al., in preparation (two-trait constraint); PSD generalisation is algebraic
- *Function*: .check_pleio_feasible

- *Location:* R/effects_pleioarch.R:870-917
- *Notes:* Multi-trait check on the variance-free matrix avoids tolerance masking

V2-grammar-C4. Total pleiotropic correlation target

$$r_{ij}^{tot} = \frac{\sum_c r_{c,ij} \sqrt{V_{c,i} V_{c,j}}}{\sqrt{\sum_c V_{c,i} \sum_c V_{c,j}}}$$

- *Variables:* c = mean-effect layers (A, D, E), $V_{c,t}$ per-layer per-trait prop
- *Reference:* Package derivation (covariance addition + Cauchy-Schwarz); no external source
- *Function:* `.pleio_total_cor_check`
- *Location:* R/effects_pleioarch.R:724-788 (roxygen 543-574)
- *Notes:* Equals r_{ij} iff per-trait variance profiles are proportional or $r_{ij} = 0$; warns when off by > 1 %

V2-grammar-C5. Additive MAF effect scaling

$$s_j = 1/\sqrt{2 \text{MAF}_j(1 - \text{MAF}_j)}$$

- *Variables:* marker MAF; $s_j \leftarrow 1$ when MAF = 0 or non-finite
- *Reference:* Claimed PleioArch `scaleQTNEffects.R`; unavailable, page unverified
- *Function:* `.pleio_draw`
- *Location:* R/effects_pleioarch.R:201-217 (144-149)
- *Notes:* P3: implements the documented factor; source parity UNVERIFIABLE

V2-grammar-C6. LD window (two-trait linked distinct loci)

$$r_{jk}^2 = \text{Cor}(X_j, X_k)^2 \in [r_{\min}^2, r_{\max}^2], \text{ same chromosome, causal loci disjoint}$$

- *Variables:* dosage columns X ; `ld_type` direct/indirect
- *Reference:* Package v2 implementation; Fernandes & Lipka 2020 *BMC Bioinformatics* 21:491 (v1 LD concept; page unverified for the v2 algorithm)
- *Function:* `.draw_qtn_ld`
- *Location:* R/arch_ld.R:62-225 (window 69-95, sampling 105-175)
- *Notes:* DECISION-014; indirect path also checks the causal pair's own r^2

Effects and architectures: PleioArch pleiotropy, effect series, independent and LD architectures

Source report: v2-effects-arch.md

Entries: 23

V2-effects-arch-1. Pleiotropic covariance

$$\Sigma_{ii} = \pi_i V_i, \quad \Sigma_{ij} = \rho_{ij} \sqrt{V_i V_j}$$

- *Variables:* π_i shared share, V_i = layer prop of trait i (phenotypic variance 1), ρ_{ij} = cor

- *Reference:* simulateEffects.R:34, 38–39 (Prado et al., in prep.; page n/a — bundled code)
- *Function:* .pleio_draw, .pleio_nonadditive_draw
- *Location:* effects_pleioarch.R:90–91, :504–505
- *Notes:* F1 bit-exact vs reference; generalised to n traits (DECISION-013)

V2-effects-arch-2. Trait-specific variance

$$V_i^{spec} = (1 - \pi_i) V_i, \text{ i.i.d. per trait}$$

- *Variables:* as above
- *Reference:* simulateEffects.R:45–46
- *Function:* .pleio_draw, .pleio_unit_effects
- *Location:* :197–199, :685
- *Notes:* zero-variance classes still drawn (EFF-F4)

V2-effects-arch-3. Major/minor split

$$\Sigma_{maj} = \Sigma \phi, \Sigma_{min} = \Sigma(1 - \phi); \text{ per-unit } \Sigma_{maj}/n_{maj}, \Sigma_{min}/n_{min}$$

- *Variables:* $\phi = \text{prop_var_major}$
- *Reference:* simulateEffects.R:49–60, 69
- *Function:* .pleio_draw, .draw_mvnorm
- *Location:* :195–196, :934
- *Notes:* additive layer only (:63–68)

V2-effects-arch-4. MVN draw

$$E = Z \Sigma_{per}^{1/2}, \Sigma_{per}^{1/2} = U \text{diag}(\sqrt{\lambda_+}) U^\top$$

- *Variables:* $Z \sim N(0, I) \ n \times n_t$
- *Reference:* reference uses chol (:73); symmetric root is standard linear algebra (page unverified)
- *Function:* .draw_mvnorm
- *Location:* :929–939
- *Notes:* same law, different realizations; handles singular Σ

V2-effects-arch-5. Univariate draw

$$e_k \sim N(0, V_i^{spec}/n_{spec})$$

- *Reference:* simulateEffects.R:78–83
- *Function:* .draw_univariate
- *Location:* :944–949

V2-effects-arch-6. Shared-unit count

$$n_{pleio} = \text{round}(\bar{\pi} n), n_{spec} = n - n_{pleio}$$

- *Variables:* $\bar{\pi}$ = mean of π_i
- *Reference:* package design (reference takes pleioSnps as input)
- *Function:* `.pleio_partition`
- *Location:* :291–292, 320
- *Notes:* R half-to-even (EFF-F9)

V2-effects-arch-7. Attainability (2 traits)

$$\rho^2 \leq \pi_1 \pi_2$$

- *Reference:* `simulateEffects.R:17`
- *Function:* `.check_pleio_feasible`
- *Location:* :876–890
- *Notes:* `tol 8\,\epsilon\max(\cdot)`

V2-effects-arch-8. Attainability (n traits)

$$M \succeq 0, M_{ii} = \pi_i, M_{ij} = \rho_{ij}; \Sigma = D^{1/2} M D^{1/2}$$

- *Variables:* $D = \text{diag}(V)$
- *Reference:* congruence preserves PSD (standard; page unverified)
- *Function:* `.check_pleio_feasible`
- *Location:* :898–915
- *Notes:* E6 catches min eig -0.2

V2-effects-arch-9. Allele -> genotype scaling

$$e_k^* = e_k / \sqrt{2 \text{MAF}_k (1 - \text{MAF}_k)}$$

- *Variables:* MAF from $p = (g + 1)/2$, $\text{MAF} = \min(p, 1 - p)$
- *Reference:* `scaleQTNEffects.R:26–37`; $\text{Var}(\text{Bin}(2, p)) = 2p(1 - p)$ (elementary)
- *Function:* `.pleio_draw`; `.marker_maf_ref` (via `.marker_stats_ref`)
- *Location:* :201–217; `grammar_simulate_phenotype.R:1040–1044`
- *Notes:* HWE sd of -1/0/1 dosage

V2-effects-arch-10. Non-additive normaliser

$$e_u^* = e_u / \widehat{\text{sd}}(z_u), z_u = \text{het indicator or } \prod_k (x_k - \bar{x}_k)$$

- *Variables:* sample sd (n-1)
- *Reference:* DECISION-023 (own design, :332)
- *Function:* `.pleio_unit_effects`; `.epi_unit_column`
- *Location:* :601–604, 686; `grammar_realize.R:579–589`; dominance design :518 vs `grammar_realize.R:510`

- *Notes:* constant columns -> effect 0, excluded from allocation : 454–457

V2-effects-arch-11. Effective per-component target

$$\rho_{ij}^{eff} = \Sigma_{ij} \mathbb{1}_{sh} / \sqrt{v_i^{eff} v_j^{eff}}, v_i^{eff} = \pi_i V_i \mathbb{1}_{sh} + (1 - \pi_i) V_i \mathbb{1}_{spec,i}$$

- *Variables:* indicators = live units exist
- *Reference:* DECISION-023 round 16
- *Function:* `.pleio_unit_effects`
- *Location:* :659–664

V2-effects-arch-12. Total-correlation target

$$\rho_{ij}^{tot} = \frac{\sum_c \rho_{ij}^{(c)} \sqrt{V_{ci} V_{cj}}}{\sqrt{\sum_c V_{ci} \sum_c V_{cj}}} \leq \rho_{ij} \text{ (Cauchy-Schwarz)}$$

- *Variables:* c over A/D/E layers
- *Reference:* DECISION-023 §3 (own derivation)
- *Function:* `.pleio_total_cor_check`
- *Location:* :754–759
- *Notes:* warns if $\text{abs}(\text{tot} - \text{cor}) > 0.01 * \text{abs}(\text{cor})$:616; E12 gives 0.400 for (.4,.1)/(.1,.4) at 0.5

V2-effects-arch-13. Layer rescale (context)

$$c_t \leftarrow (c_t - \bar{c}_t) \sqrt{\text{prop}_t} / \widehat{\text{sd}}(c_t)$$

- *Reference:* SPEC §2
- *Function:* `.genetic_matrix`
- *Location:* `grammar_realize.R`:144–146
- *Notes:* makes realized r a random ratio (P1 wording)

V2-effects-arch-14. Few-unit attenuation

$$\mathbb{E}[r] / \rho \approx 0.66, 0.82, 0.92, 0.96, 0.98, 0.99 \text{ at } K = 1, 2, 5, 10, 20, 60 (\rho = .5, \pi = 1, \text{orthogonal designs})$$

- *Variables:* K shared units
- *Reference:* numerical (F2, 40 000 draws); no closed form cited
- *Function:* doc claim
- *Location:* :26–28; `grammar_simulate_phenotype.R`:318–320
- *Notes:* verified

V2-effects-arch-15. Complete-LD ensemble mean

$$\mathbb{E}[r] = 2 \arcsin(\rho) / \pi \text{ from } P(XY > 0) = \frac{1}{2} + \arcsin(\rho) / \pi$$

- *Variables:* bivariate normal orthant

- *Reference*: Sheppard's orthant formula (page unverified)
- *Function*: doc claim
- *Location*: :28–30; DECISIONS.md 023 Scope
- *Notes*: 1/3 at $p = .5$; E7 realized ± 1

V2-effects-arch-16. Geometric effect series

$a_k = b^k$, $k = 1..n$ (default $b = 0.5$)

- *Reference*: v1 sim_method = "geometric" (Fernandes & Lipka 2020; page unverified)
- *Function*: .effect_series
- *Location*: effects_series.R:55, 64
- *Notes*: custom vector used verbatim :32–38

V2-effects-arch-17. Repulsion phase

$a_k \leftarrow a_k(-1)^{k+1}$, same for every trait

- *Reference*: package design
- *Function*: .apply_phase
- *Location*: grammar_layers.R:1482–1490
- *Notes*: applied to PleioArch effects too (EFF-F11)

V2-effects-arch-18. Residual draw

$e \sim N(0, \sigma_e^2)$, then $e \leftarrow (e - \bar{e})\sigma_e / \widehat{\text{sd}}(e)$, $\sigma_e^2 = 1 - \sum \text{prop}$

- *Reference*: SPEC §2
- *Function*: .draw_residual
- *Location*: effects_series.R:85–102
- *Notes*: exact-variance residual under the default residual_mode = "fixed"; "random" skips the rescaling (:94)

V2-effects-arch-19. LD measure

$r^2 = \text{cor}(g_a, g_b)^2$ on -1/0/1 dosage (composite)

- *Variables*: same-chromosome candidates
- *Reference*: standard Pearson (genotypic/"composite" r^2 ; page unverified)
- *Function*: window_partners, r2_pair
- *Location*: arch_ld.R:111, 124–126
- *Notes*: inclusive window :82

V2-effects-arch-20. Direct pair choice

$t_2 = \arg \max_{j \in W(t_1)} r_{t_1 j}^2$ (default partner = "strongest"); $t_2 \sim U\{W(t_1)\}$ for partner = "random"

- *Variables:* W = in-window unused partners
- *Reference:* package design (undocumented)
- *Function:* `.draw_qtn_ld`
- *Location:* `arch_ld.R:149–160 (:151–152); partner :73–74`
- *Notes:* EFF-F2

V2-effects-arch-21. Indirect flanks

$\text{pos}(u) < \text{pos}(c) < \text{pos}(d), r_{uc}^2, r_{dc}^2, r_{ud}^2 \in [r_{min}^2, r_{max}^2]$, strongest-to-cause first

- *Variables:* c = hidden cause
- *Reference:* DECISION-014
- *Function:* `.draw_qtn_ld`
- *Location:* `arch_ld.R:161–197`
- *Notes:* reported $r^2 = r_{ud}^2 : 170–174$ (EFF-F7)

V2-effects-arch-22. Layer sub-seed

$s = (1009 \text{ seed} + 7919 \sum \text{utf8}(\text{type}) + 104729 \text{ occ}) \bmod (2^{31} - 1)$

- *Reference:* package design
- *Function:* `.layer_seed`
- *Location:* `grammar_simulate_phenotype.R:1083–1093`
- *Notes:* permutation-invariant (EFF-F1)

V2-effects-arch-23. Gabriel blocks tag

keep $\arg \max_{m \in B} \text{MAF}_m$ per block B ; MAF floor 0.05; window $\lfloor 1000 \text{ kb} (1 + \epsilon) \rfloor$ bp

- *Reference:* Gabriel et al. 2002 Science 296:2225–2229 (page unverified); PLINK 1.9 constants (unverified)
- *Function:* `.gabriel_blocks`
- *Location:* `qc_ld_methods.R:24–29, 38–43`
- *Notes:* UNVERIFIABLE parity

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-effects-arch-C1. Zero-variance correlation guard

$V_i = 0 \Rightarrow \text{Cor}(g_i, g_j) = 0/0$ undefined

- *Variables:* $V_i = \text{vg}[i]$ (layer prop), R_{ij} = requested cor
- *Reference:* Definition of correlation; no external source
- *Function:* `.pleio_check_zero_var()`
- *Location:* `R/effects_pleioarch.R:246–275`

- *Notes:* nonzero requested pair with a zero-variance trait -> error (:186); zero requested pair -> warning, reported NA (:193)

V2-effects-arch-C2. Shared/specific unit count (clamped)

$$n_{sh} = \min(n, \max(0, \text{round}(n \bar{\pi}))), \quad n_{sp} = n - n_{sh}$$

- *Variables:* $n = n_qtn/n_pairs$; $\bar{\pi} = \text{mean}(\pi)$
- *Reference:* Package generalisation (reference takes `pleioSnps` as input)
- *Function:* `.pleio_partition()`
- *Location:* `R/effects_pleioarch.R:288–329` (clamp :291–292)
- *Notes:* refines Fable's row (which omits the clamp); errors when a requested class rounds to zero (:241–256); R half-to-even (EFF-F9)

V2-effects-arch-C3. One shared unit

$$g_i = a_i z \Rightarrow |\text{Cor}(g_i, g_j)| = 1 \text{ when neither trait has specific variance; otherwise one noisy draw around cor}$$

- *Variables:* z = the single shared design column; a_i = trait- i effect
- *Reference:* Algebraic consequence; no external source
- *Function:* `.pleio_single_unit_consequence()`
- *Location:* `R/effects_pleioarch.R:412–441`
- *Notes:* pairs with $|R_{ij}| \geq 1$ or zero variance skipped (:277–278); feeds the warning at :235–241

V2-effects-arch-C4. Correlation input expansion

$$\text{scalar } r \mapsto R, \quad R_{ii} = 1, \quad R_{ij} = r; \text{ matrix must be } n_t \times n_t, \text{ finite, in } [-1, 1], \text{ symmetric, unit diagonal}$$

- *Variables:* `cor, n_traits`
- *Reference:* Correlation-matrix definition; no external source
- *Function:* `.pleio_cor_matrix()`
- *Location:* `R/effects_pleioarch.R:957–986`
- *Notes:* PSD is checked later by `.check_pleio_feasible()` after the diagonal is replaced by π_i

V2-effects-arch-C5. Pleiotropic-share argument mapping

$$\pi = (\pi_T, \pi_S) \text{ from } \pi_target/\pi_secondary, \text{ or scalar/length-}n_t \text{ } \pi; \text{ all in } [0, 1]$$

- *Variables:* `pi, pi_target, pi_secondary`
- *Reference:* Bundled PleioArch interface (`simulateEffects.R:10: piT, piS`); Prado et al. in prep., page unverified
- *Function:* `.pleio_pi_vector()`
- *Location:* `R/effects_pleioarch.R:829–860`
- *Notes:* spellings mutually exclusive (:684–686); two-trait spelling rejected for $n_t > 2$ (:689–691)

V2-effects-arch-C6. Architecture-specific QTN sampling

independent: per-trait SRSWOR from C ; pleiotropy: one shared SRSWOR replicated to every trait; ld: delegated to `.draw_qtn_ld()`

- *Variables:* C = candidate markers
- *Reference:* Fernandes & Lipka 2020 BMC Bioinf 21:491 (page unverified); package v2 design
- *Function:* `.draw_qtn()`
- *Location:* `R/arch_independent.R:21–45`
- *Notes:* sub-seed set and restored (: 27–31); the pleiotropy branch : 34–37 is reached only by `vqtl(same_as_add = FALSE)` (EFF-F8) — the additive PleioArch path uses `.pleio_draw()`

V2-effects-arch-C7. Distinct-chromosome allocation

chromosome $k \mapsto 1 + ((k - 1) \bmod T)$ (first-occurrence order), then SRSWOR within the assigned pool

- *Variables:* $T = n_{\text{traits}}$
- *Reference:* Package design; no external source
- *Function:* `.draw_qtn_distinct_chr()`
- *Location:* `R/arch_independent.R:56–75`
- *Notes:* errors if fewer than T chromosomes (: 60–63) or a pool is short (: 68–72)

V2-effects-arch-C8. Epistatic-set sampling

draw $n_{\text{pairs}} \times$ interaction markers SRSWOR, reshape by row

- *Variables:* candidate markers; interaction
- *Reference:* Package design; no external source
- *Function:* `.draw_qtn_pairs()`
- *Location:* `R/arch_independent.R:81–103`
- *Notes:* pleiotropy branch : 98–101 unreachable (EFF-F8); PleioArch non-additive uses `.pleio_units()`

V2-effects-arch-C9. Candidate-marker rule

$$C = \{j : \text{isfinite}(\text{MAF}_j) \wedge \text{MAF}_j > 0\}$$

- *Variables:* `sim$maf`
- *Reference:* Package design
- *Function:* `.candidate_markers()`
- *Location:* `R/arch_independent.R:114–134`
- *Notes:* makes the `s == 0` guard at `effects_pleioarch.R:205` dead

V2-effects-arch-C10. Layer occurrence index

$$o_t = \sum_l \mathbb{1}(\text{type}_l = t)$$

- *Variables:* prior layers, requested type
- *Reference:* Package RNG design (grammar seed contract)
- *Function:* `.type_occurrence()`
- *Location:* `R/arch_independent.R:139–141`
- *Notes:* feeds `.layer_seed()`; type-local, so inserting a different layer type does not perturb an additive draw

Crossing: populations, meiosis draws, maps, pedigree, mating designs, crossbreeding

Source report: v2-crossing.md

Entries: 24

V2-crossing-1. Heterozygote phasing

$$\text{cis}_j = \mathbf{1}(x_j \geq 0), \text{trans}_j = \mathbf{1}(x_j > 0)$$

- *Variables:* $x_j \in \{-1, 0, 1\}$ dosage
- *Reference:* package convention (documented assumption)
- *Function:* `as_population`
- *Location:* `R/cross_population.R:176–177`
- *Notes:* every het $\rightarrow$ allele 1 on cis (coupling)

V2-crossing-2. Dosage from strands

$$x_j = \text{cis}_j + \text{trans}_j - 1$$

- *Variables:* $\text{strands} \in \{0, 1\}$
- *Reference:* `isqg genotype_num (Genetics.cpp:363)`
- *Function:* `dosages`
- *Location:* `R/cross_population.R:623; genome.rs:273–278`
- *Notes:* integer output

V2-crossing-3. Crossover count

Default (`interference = NULL`, since 2.0.0.9003): two-pathway gamma model with $\nu = 2.6$, $p = 0$. Chiasmata on the bivalent at 2 per Morgan; a share p is a Poisson process (drawn at the gamete level, $n \sim \text{Poisson}(pL)$, positions $U(0, L)$); the rest is a stationary renewal process with gaps $\sim \text{Gamma}(\nu, 2\nu(1 - p))$, first point $U \cdot \text{Gamma}(\nu + 1, 2\nu(1 - p))$, each chiasma kept in the gamete with probability $1/2$. `interference = "poisson"` (the default before 2.0.0.9003, `isqg stream`): $n_x \sim \text{Poisson}(L)$. $L = \text{last cM}/100$ in both

- *Variables:* L in Morgans
- *Reference:* Poisson branch: Karlin & Liberman 1978 PNAS 75:6332–6336 (as cited by `isqg`; not re-verified); `isqg Genetics.cpp:84`. Gamma model: McPeck & Speed 1995 *Genetics* 139(2):1031–1044; two-pathway extension Housworth & Stahl 2003 *Am. J. Hum. Genet.* 73(1):188–197 (as cited in `man/cross.Rd`; pages not verified here)
- *Function:* `.draw_meiosis`, `.draw_meiosis_interference`, `.check_interference`
- *Location:* `R/cross_mating.R:53–56 (dispatch)`, Poisson branch : 77–78; gamma model : 215–275; default `.INTERFERENCE_DEFAULT : 117 (DECISION-047)`; `cM  $\rightarrow$  M : 339, 341`
- *Notes:* last position, not span (CROSS-F5), in both models; seeded meiosis results changed with the new default (NEWS, 2.0.0.9003)

V2-crossing-4. Chiasma positions

$$u_1, \dots, u_{n_x} \stackrel{iid}{\sim} U(0, L), \text{ sorted}$$

- *Reference:* `isqg Genetics.cpp:93–95`

- *Function:* `.draw_meiosis`
- *Location:* `R/cross_mating.R:79–87`
- *Notes:* not drawn when $n_x = 0$; Poisson branch only (`interference = "poisson"`); default gamma model in V2-crossing-3

V2-crossing-5. Strand choice

$f \sim \text{Bernoulli}(1/2)$, always drawn

- *Reference:* `isqg Genetics.cpp:74`
- *Function:* `.draw_meiosis`
- *Location:* `R/cross_mating.R:90`
- *Notes:* unconditional; the gamma branch also draws one flip per slot (:272)

V2-crossing-6. Ancestry mask

$$m_j = f \oplus \bigoplus_k \mathbf{1}(j \geq b_k), \quad b_k = \#\{p_i \leq u_k\}$$

- *Variables:* p_i marker positions
- *Reference:* `isqg Genetics.cpp:65-69, 74-75`
- *Function:* `chromosome_mask, breaks_at`
- *Location:* `src/rust/src/meiosis.rs:49–51, 63–72`
- *Notes:* $\leq$ (`upper_bound`)

V2-crossing-7. Gamete

$$g_j = m_j \text{ cis}_j + (1 - m_j) \text{ trans}_j$$

- *Reference:* `isqg Genetics.cpp:356`
- *Function:* `recombine`
- *Location:* `meiosis.rs:224–237`

V2-crossing-8. Cross / self / DH

`cross: (cis,trans) = (g{(1)}{p1}, g{(2)}{p2}); self: same with  $p_1 = p_2$ , 2 independent meioses; DH: (g, g)`

- *Reference:* `isqg Mating.cpp:91-95, 112-116, 127-130; Genetics.cpp:521-541`
- *Function:* `mate_haplotypes`
- *Location:* `meiosis.rs:350–371; events per progeny R/cross_mating.R:345`
- *Notes:* progeny-major, parent-1 then parent-2

V2-crossing-9. Recombination fraction (implied, verified)

$$r = \frac{1}{2} (1 - e^{-2d})$$

- *Variables:* d Morgans
- *Reference:* Haldane 1919 J. Genet. 8:299–309 (page unverified; standard)

- *Function*: — (property of the Poisson process; holds for interference = "poisson", or $\nu = 1/p = 1$, not for the default gamma model, whose $r(d) = [1 - P_0(d)]/2$ is given in `man/cross.Rd`)
- *Location*: `verified e4.R/e5.R; tests/testthat/test-cross.R:179–202` (file runs under `simplePHENOTYPES.interference = "poisson", :13`)
- *Notes*: holds for any map origin

V2-crossing-10. Heterozygosity under selfing

$$H_t = H_0/2^t$$

- *Reference*: Falconer & Mackay 1996 ch. 5 (page unverified)
- *Function*: `selfcross` (doc)
- *Location*: `R/cross_mating.R:627–629; verified e1.R E3`

V2-crossing-11. Synthetic map rate

$$r(p) = 1 - s \exp\left(-\frac{1}{2} \left(\frac{p-c}{wL}\right)^2\right)$$

- *Variables*: s suppression, w width, c centromere, L bp span
- *Reference*: phenomenological (package); scale $\approx$ Bauer et al. 2013 GB 14:R103 (UNVERIFIABLE numbers)
- *Function*: `synthetic_map`
- *Location*: `R/cross_map.R:157`

V2-crossing-12. Synthetic map integration

$$cM_i = \frac{\sum_{k < i} (p_{k+1} - p_k) \frac{r_k + r_{k+1}}{2}}{\sum_k (\cdot)} \cdot \text{len}_{cM}; \text{len}_{cM} = cM/\text{Mb} \cdot L/10^6 \text{ if total_cm NULL}$$

- *Reference*: package
- *Function*: `synthetic_map`
- *Location*: `R/cross_map.R:168–173, :133`
- *Notes*: trapezoid; monotone

V2-crossing-13. Additive value (fixed scale)

$$A_i = \sum_j x_{ij} \beta_j$$

- *Reference*: Falconer & Mackay 1996 (a on $-1/0/1$ scale; page unverified)
- *Function*: `additive_value`
- *Location*: `R/cross_population.R:746`
- *Notes*: no centring

V2-crossing-14. Genotypic value

$$G_i = \sum_j [a_j x_{ij} + d_j \mathbf{1}(x_{ij} = 0)]$$

- *Reference*: Falconer & Mackay 1996 ch. 7 (page unverified)

- *Function*: genotypic_value
- *Location*: R/cross_population.R:821

V2-crossing-15. Residual from h^2

$$\sigma_e^2 = \text{Var}(g_{\text{ref}}) (1 - h^2) / h^2$$

- *Reference*: $h^2 = V_A / (V_A + V_E)$, F&M 1996 (page unverified)
- *Function*: phenotype_value
- *Location*: R/cross_population.R:1061

V2-crossing-16. Breed composition

$$F_{i,b} = \frac{1}{2}(F_{m(i),b} + F_{f(i),b}), \text{ founders } F = \mathbf{1}(\text{pool} = b)$$

- *Reference*: SPEC-block3b §7 (package)
- *Function*: breed_composition
- *Location*: R/cross_breed.R:37–43
- *Notes*: expected, not realized

V2-crossing-17. Realized heterosis

$$H = \bar{G}_{\text{cross}} - \sum_b \bar{F}_b \bar{G}_b$$

- *Variables*: $\bar{F}_b$ mean composition
- *Reference*: mid-parent heterosis, F&M 1996 ch. 14 (page unverified)
- *Function*: heterosis
- *Location*: R/cross_breed.R:135–139, 152
- *Notes*: composition-weighted

V2-crossing-18. Expected F1 heterosis

$$H_{F1} = \sum_j d_j [h_{AB,j} - \frac{1}{2}(h_{A,j} + h_{B,j})], h_{AB} = p_A(1 - p_B) + (1 - p_A)p_B; \text{HWE: } \sum_j d_j (p_{Aj} - p_{Bj})^2;$$

inbred lines: $\sum_j d_j (p_A + p_B - 2p_A p_B)$

- *Variables*: p = allele-1 gamete freq
- *Reference*: F&M 1996 $H_{F1} = \sum dy^2$ (page unverified); general form = package derivation
- *Function*: heterosis + .expected_cross_means
- *Location*: R/cross_breed.R:140–150; R/select_combining.R:244–255
- *Notes*: verified exactly (E12)

V2-crossing-19. Per-locus cross mean

$$E = a(g_i + g_k - 1) + d(g_i + g_k - 2g_i g_k)$$

- *Reference*: expansion of $p_{AA}a + p_{Aa}d - p_{aa}a$

- *Function:* `.expected_cross_means`
- *Location:* `R/select_combining.R`:249–254

V2-crossing-20. Retention fractions (doc only)

F2, BC: $1/2 H_{F1}$; 2-breed rotation: $2/3$; n-breed: $(2^n - 2)/(2^n - 1)$

- *Reference:* SPEC §7 derivation (breed-origin dominance model), **requires within-breed HWE**
- *Function:* `roxygen`
- *Location:* `R/cross_breed.R`:68–72; `docs/SPEC-block3b.md` §7
- *Notes:* CROSS-F1

V2-crossing-21. Rotation sire sequence

$\text{sire}_g = \text{breed}((g - 1) \bmod n_b) + 1$

- *Reference:* —
- *Function:* `crossbreed`
- *Location:* `R/cross_breed.R`:302–305

V2-crossing-22. Pedigree key

$\text{key} = \text{FNV-1a-128}(\text{canonical}(\text{design}, \text{keys}_{p_1}, \text{keys}_{p_2}, \text{RNG state before/after, counts, flips, chiasma bits, n})) \parallel \text{"_i"}$

- *Reference:* FNV spec (128-bit params)
- *Function:* `.mating_pedigree`, `.stable_key`
- *Location:* `R/cross_pedigree.R`:25–37 (`.stable_key`), :42–59 (`.key_part`), :205–207 (mating key); `hash.rs`:12–23
- *Notes:* founder key :97–138 (`.founder_pedigree`, batched since the post-audit perf change)

V2-crossing-23. Generation

$\text{gen}_i = 1 + \max(\text{gen}_m, \text{gen}_f)$

- *Reference:* —
- *Function:* `.mating_pedigree`
- *Location:* `R/cross_pedigree.R`:208–209

V2-crossing-24. A-matrix (other group, consistency only)

$A_{ij} = \frac{1}{2}(A_{i,m(j)} + A_{i,f(j)}), A_{jj} = 1 + \frac{1}{2}A_{m(j)f(j)}, \text{DH: } 2$

- *Reference:* Wright 1922; Emik & Terrill 1949 (pages as cited in `roxygen`, not re-verified)
- *Function:* `a_matrix`
- *Location:* `R/select_blup.R`:69–90
- *Notes:* consistent with this group's pedigree (E10)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-crossing-C1. Average-effect vs genotypic-value distinction (doc)

$$\alpha_j = a_j + d_j(1 - 2p_j), \quad A_i = \sum_j \alpha_j(x_{ij} - 2p_j)$$

- *Variables:* p_j counted-allele frequency, x_{ij} gene count
- *Reference:* Falconer & Mackay 1996 (secondary; page unverified); Fisher 1918 average effect (not cited in roxygen)
- *Function:* roxygen of genotypic_value (implemented in select_ind(on = "bv"), other group)
- *Location:* R/cross_population.R:774–780 (verified: a text at 776-777)
- *Notes:* Doc correctly states additive_value(alpha) is only ranking-equivalent (differs by an additive constant)

V2-crossing-C2. Full-sib family identity

fam(i) = $\{k_{m(i)}, k_{f(i)}\}$ (unordered key pair); maternal/paternal half-sib = k_m / k_f ; selfed = k_m iff design $\in \{\text{self, dh}\}$

- *Variables:* stable parent keys k
- *Reference:* package bookkeeping (no external source needed)
- *Function:* families
- *Location:* R/cross_pedigree.R:307–336 (verified: families <- at 307, closes at 336)
- *Notes:* ifelse(m <= f, paste(m,f), paste(f,m)) merges reciprocals; labels made unique

V2-crossing-C3. Mating designs (counts)

factorial $|M| \times |F|$ (minus selfs); diallel $n(n-1)$ (+ n with selfs); half-diallel $\binom{n}{2}$ (+ n); nested: m distinct mothers per father via bipartite matching; random: mother uniform, then admissible father uniform

- *Variables:* parent keys, progeny_per_cross, allow_self
- *Reference:* Griffing 1956 Aust. J. Biol. Sci. 9:463–493 for the diallel methods (standard citation, pages not verified here; not cited in roxygen); NC designs I/II uncited
- *Function:* mating_design
- *Location:* R/cross_mate.R:360–486 (verified: mating_design <- at 229; file is 350 lines)
- *Notes:* Fable's rubric line covers the counts (e2.R E13) but Fable's equation map omitted the row

V2-crossing-C4. Within-family relationship validation (test)

expected BV correlations: S1 sibs $2/3$ ($A_{ij} = 1, A_{ii} = 1.5$), full sibs and DH $1/2$; ICC = $\frac{MS_B - MS_W}{MS_B + (k-1)MS_W}$

- *Variables:* k balanced family size, MS_B, MS_W
- *Reference:* Wright 1922 (relationship; page unverified); one-way ANOVA ICC estimator (Fisher 1925 / standard; uncited)
- *Function:* test helper .icc
- *Location:* tests/testthat/test-family-relationship.R:1–58 (verified: 58 lines, .icc at 26, targets at 52-57)
- *Notes:* validates the crossing engine's realized relationships, not S3 index weights

Rust core: count-location meiosis, genome masks, numericalization (isqg parity)

Source report: v2-rust-core.md

Entries: 16

V2-rust-core-1. Crossover count (count-location)

$n_x \sim \text{Poisson}(L)$ (interference = "poisson" only; the default gamma model is V2-crossing-3)

- *Variables:* L = last map position of the chromosome, Morgans
- *Reference:* Karlin & Liberman 1978, PNAS 75(12):6332–6336 (as cited in isqg Genetics.cpp:79; page unverified by me)
- *Function:* .draw_meiosis
- *Location:* R R/cross_mating.R:77–78; isqg Genetics.cpp:44,84; Rust: none (input counts)
- *Notes:* L is NOT the span (last – first); RUST-F10 docstring

V2-rust-core-2. Chiasma locations

$x_i \stackrel{iid}{\sim} U(0, L)$, $i = 1..n_x$, sorted

- *Variables:* x_i
- *Reference:* Karlin & Liberman 1978 (as above)
- *Function:* .draw_meiosis
- *Location:* R :79–87; isqg Genetics.cpp:86–97
- *Notes:* R runif is on the open interval; Rust never sorts (meiosis.rs:60–62)

V2-rust-core-3. Strand raffle

$f \sim \text{Bernoulli}(1/2)$, drawn unconditionally

- *Variables:* f
- *Reference:* isqg convention (Genetics.cpp:74)
- *Function:* .draw_meiosis / chromosome_mask
- *Location:* R :90; Rust meiosis.rs:68–70; isqg :74–75
- *Notes:* skipping it when $n_x = 0$ desynchronises the stream

V2-rust-core-4. Breakpoint rank

$b(x) = \#\{j : p_j \leq x\}$

- *Variables:* p_j ascending map positions
- *Reference:* std::upper_bound semantics (isqg)
- *Function:* breaks_at
- *Location:* Rust meiosis.rs:49–51; isqg Genetics.cpp:67
- *Notes:* 0-based; $b = n \Rightarrow$ no-op; marker exactly at x stays upstream

V2-rust-core-5. Ancestry mask

$$m_j = f \oplus \bigoplus_i \mathbb{1}[j \geq b(x_i)]$$

- *Variables:* $m_j \in \{0, 1\}$
- *Reference:* isqg Genetics.cpp:54–77
- *Function:* chromosome_mask, Bits::toggle_from, flip_all
- *Location:* Rust meiosis.rs:63–72, genome.rs:69–87; isqg :58–70 (XOR loop), :74–75 (flip)
- *Notes:* duplicates cancel; order irrelevant

V2-rust-core-6. Gamete assembly

$$g_j = m_j c_j + (1 - m_j) t_j$$

- *Variables:* c, t = parental cis/trans strands
- *Reference:* isqg DNA::recombination
- *Function:* recombine
- *Location:* Rust meiosis.rs:224–237; isqg Genetics.cpp:356

V2-rust-core-7. Cross

$$\text{progeny } i: (\text{cis}, \text{trans}) = (g_{2i}^{(P_1)}, g_{2i+1}^{(P_2)})$$

- *Variables:* event index progeny-major
- *Reference:* isqg Mating.cpp:92–94
- *Function:* mate_haplotypes
- *Location:* Rust meiosis.rs:359–369; R .mate cross_mating.R:345, 453, 460–467
- *Notes:* cis from parent 1 (tested by cross_swapped)

V2-rust-core-8. Self

as cross with $P_2 = P_1$, two independent meioses

- *Reference:* isqg Mating.cpp:113–115
- *Function:* selfcross $\rightarrow$.mate(parent, parent)
- *Location:* R cross_mating.R:647–652; Rust same path

V2-rust-core-9. Doubled haploid

$$(\text{cis}, \text{trans}) = (g_i^{(P)}, g_i^{(P)})$$

- *Variables:* one event per progeny
- *Reference:* isqg Mating.cpp:128–129 $\rightarrow$ duplication Genetics.cpp:532–541
- *Function:* mate_haplotypes (Dh)
- *Location:* Rust meiosis.rs:263–269, 351–358; R cross_mating.R:686–691

V2-rust-core-10. Genotype projection

$$G_j = \begin{cases} 1 & c_j \wedge t_j \\ 0 & c_j \oplus t_j \\ -1 & \neg c_j \wedge \neg t_j \end{cases}$$

- *Reference:* isqg DNA::genotype_num Genetics.cpp:358–369
- *Function:* write_genotype, dosages
- *Location:* Rust genome.rs:273–284; R cross_population.R:623 (cis + trans – 1)
- *Notes:* lossy (phase)

V2-rust-core-11. Founder phasing

$$c_j = \mathbb{1}[d_j \geq 0], \quad t_j = \mathbb{1}[d_j > 0]$$

- *Variables:* $d_j \in \{-1, 0, 1\}$ dosage
- *Reference:* isqg founder("Aa") Genetics.cpp:307–309
- *Function:* as_population
- *Location:* R cross_population.R:176–177
- *Notes:* all hets coupling-phased (RUST-F11)

V2-rust-core-12. Map units

$$p_j = \text{cM}_j / 100$$

- *Reference:* definition
- *Function:* .mate
- *Location:* R cross_mating.R:339, 341

V2-rust-core-13. Haldane map function (consequence, tested only)

$$r(d) = \frac{1}{2} (1 - e^{-2d})$$

- *Variables:* d Morgans between two markers
- *Reference:* Haldane 1919, J. Genet. 8:299–309 (page unverified)
- *Function:* —
- *Location:* test tests/testthat/test-cross.R:179–202; auditor T2–T4
- *Notes:* follows from a rate-1 Poisson process with no interference (interference = "poisson"; not the default gamma model)

V2-rust-core-14. Numericalization — orientation

$$\text{allele-2 major} \iff n_2 > n_0 \iff 2n_2 + n_1 > 2n_0 + n_1$$

- *Variables:* n_k = count of raw dosage k
- *Reference:* package contract (io_as_numeric.R:8–12)
- *Function:* compute_flip

- *Location:* R `io_detect_format.R`:362–364; reference mode :354–359
- *Notes:* strict >: tie → allele-1 (RUST-F9)

V2-rust-core-15. Numericalization — codes

–101: (major, het, minor) = (1,0,–1); 012: (2,1,0); Dom: non-het→minor; Left: het→minor; Right: het→major; impute Middle/Minor/Major → het/minor/major class

- *Reference:* package contract; v1 `numericalization.R` (Dom/Left/Right lines)
- *Function:* `numericalize_core`
- *Location:* Rust `numeric.rs`:76–88,135–180
- *Notes:* model applied after imputation (fixes a v1 gap, `numeric.rs`:127–134)

V2-rust-core-16. Pedigree key hash

$$h_0 = \text{offset}_{128}; h_{k+1} = (h_k \oplus b_k) \cdot p \bmod 2^{128}, p = 2^{88} + 2^8 + 0x3b$$

- *Variables:* b_k UTF-8 bytes
- *Reference:* Fowler–Noll–Vo FNV-1a (IETF draft-eastlake-fnv; section unverified)
- *Function:* `fnv1a_128`, `.stable_key`
- *Location:* Rust `hash.rs`:12–23; R `cross_pedigree.R`:22–59
- *Notes:* verified vs Python (RUST-F15)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-rust-core-C1. Public ingestion dispatch

`as_numeric(x) = format_conversion(x, to = "numeric")` after path/object disambiguation

- *Variables:* x : bare character scalar path, or in-memory genotype object (character matrix, `data.frame`, ...)
- *Reference:* Package API contract; no external theory source
- *Function:* `as_numeric()`
- *Location:* R `io_as_numeric.R`:192–207 (verified: `if (is.character(x) && is.null(dim(x)))` at:198, `format_conversion(...)` at:206)
- *Notes:* Deterministic; character matrices deliberately not treated as paths (:101–104); F4 lives downstream in `io_detect_format.R`:286–297

V2-rust-core-C2. Genetic-model post-transform (split out of Fable’s single “codes” row)

Dom: $z = c_H$ if het else c_m ; Left: het → c_m ; Right: het → c_M ; Add: unchanged

- *Variables:* z : output code; c_M, c_H, c_m : coding-specific major/het/minor values
- *Reference:* Package-defined simulation coding (v1 `numericalization.R` Dom/Left/Right lines); no external theory source
- *Function:* `numericalize_core()`
- *Location:* `src/rust/src/numeric.rs`:149–180 (verified: `match model` at:156, "Dom" :157, "Left" :164, "Right" :171, default :178)
- *Notes:* Applied *after* imputation (:64–71); Codex’s audit-only 64-case grid passed; unknown model silently = Add (F7/RC-04)

Selection engine: truncation, indices (Smith-Hazel, Lush, QGSI), culling, schemes

Source report: v2-selection.md

Entries: 18

V2-selection-1. Truncation count from prop

$$k = \max(1, \text{round}(pN))$$

- Variables: p prop, N candidates
- Reference: — (implementation)
- Function: `.resolve_keep`
- Location: R/select_ind.R:574
- Notes: R half-to-even

V2-selection-2. Selection intensity (large N)

$$i(p) = \frac{\varphi(\Phi^{-1}(1-p))}{p}$$

- Variables: p = k/N
- Reference: Falconer & Mackay 1996 (page unverified)
- Function: `.resolve_keep`
- Location: R/select_ind.R:585–587
- Notes: count with nearest i; infinite-N form

V2-selection-3. Selection differential

$$S = \bar{y}_{sel} - \bar{y}$$

- Variables: y ranking criterion (sign-flipped for low)
- Reference: Falconer & Mackay 1996 (page unverified)
- Function: `select_ind`
- Location: R/select_ind.R:460, 482
- Notes: natural sign restored for low

V2-selection-4. Realized intensity

$$i = S / \hat{\sigma}_y, \hat{\sigma} \text{ with } n-1$$

- Reference: Falconer & Mackay 1996 (page unverified)
- Function: `select_ind`, `.select_culling`
- Location: R/select_ind.R:461–478, 1089–1090
- Notes: 0 when sd = 0

V2-selection-5. Response (documented, tested here)

$$R = i h^2 \sigma_P = h^2 S$$

- *Variables*: h^2 narrow-sense
- *Reference*: Falconer & Mackay 1996 (page unverified)
- *Function*: roxygen only
- *Location*: R/select_ind.R:16–21
- *Notes*: verified numerically (exp4)

V2-selection-6. Average effect

$$\alpha_j = a_j + d_j(q_j - p_j) = a_j + d_j(1 - 2p_j)$$

- *Variables*: a, d per-locus (realized scale), p counted-allele freq
- *Reference*: Fisher 1918 52:399–433; Falconer & Mackay 1996 (page unverified)
- *Function*: .avg_effect
- *Location*: R/grammar_realize.R:182–184
- *Notes*: DECISION-019

V2-selection-7. Breeding value

$$A_i = \sum_j \alpha_j(x_{ij} - 2p_j)$$

- *Variables*: x gene content 0/1/2
- *Reference*: as above
- *Function*: .breeding_value_matrix
- *Location*: R/grammar_realize.R:236–305 (loop 282–303)
- *Notes*: refused under epistasis 250–263

V2-selection-8. Smith–Hazel index

$$b = P^{-1}Ga, I = b'y$$

- *Variables*: P = sample Cov(y), G = sample Cov(A), a economic
- *Reference*: Smith 1936 7:240–250; Hazel 1943 28:476–490
- *Function*: .index_score, .index_weights
- *Location*: R/select_ind.R:769–782; 799–842
- *Notes*: SVD pseudo-inverse if P singular; $G \neq \text{Cov}(y, A)$ off HWE (F4)

V2-selection-9. QGSI

$$\hat{I} = w'\hat{\gamma} + \hat{\gamma}'W\hat{\gamma}, W \leftarrow (W + W')/2$$

- *Variables*: $\hat{\gamma}$ = true BV per trait
- *Reference*: Ceron-Rojas et al. 2026 (art. no. unverified)
- *Function*: .quadratic_index_score
- *Location*: R/select_ind.R:737, 744–747

- *Notes:* true BV, not GEBV

V2-selection-10. Lush combined index

$$b = V^{-1}c; b_1 = \frac{h^2(1-r)}{1-rh^2}, b_2 = \frac{h^2nr(1-h^2)}{[1+(n-1)rh^2](1-rh^2)}, I = b_1(y - \bar{y}) + b_2(\bar{y}_f - \bar{y})$$

- *Variables:* $V = v_P \begin{pmatrix} 1 & m \\ m & m \end{pmatrix}$, $m = \frac{1+(n-1)t}{n}$, $c = h^2v_P(1, \frac{1+(n-1)r}{n})'$, $t = rh^2$
- *Reference:* Lush 1947 81:241–261, 362–379 (verified); Hazel 1943 (deviations)
- *Function:* `.combined_score`
- *Location:* R/select_ind.R:867, 872–877, 889–902
- *Notes:* family of one / $t = 1 \rightarrow h^2 \cdot \text{dev}$ (890–892)

V2-selection-11. Within-family allocation

largest-remainder of $k n_f / N$ capped at n_f

- *Reference:* — (implementation)
- *Function:* `.sel_within_family`
- *Location:* R/select_ind.R:939–949
- *Notes:* tie by label order

V2-selection-12. Among-family

keep whole families by $\bar{y}_f$ until $\geq k$

- *Reference:* —
- *Function:* `.sel_among_family`
- *Location:* R/select_ind.R:968–978
- *Notes:* overshoots k

V2-selection-13. Independent culling

keep $\bigcap_t \text{top}_{\lceil c_t N \rceil}(y_t)$; sequential: nested

- *Variables:* `c_t` per-trait proportions
- *Reference:* Hazel & Lush 1942 33(11):393–399 (verified)
- *Function:* `.select_culling`
- *Location:* R/select_ind.R:1067–1081
- *Notes:* DECISION-028

V2-selection-14. Index : culling : tandem

$$\sqrt{T} i(p) : T i(p^{1/T}) : i(p)$$

- *Variables:* T traits, p overall

- *Reference:* Hazel & Lush 1942 + package derivation
- *Function:* roxygen; test-culling.R:71–89
- *Location:* R/select_ind.R:157–164
- *Notes:* 1 : 0.907 : 0.707 reproduced

V2-selection-15. Bulk pool

$(n_1, \dots, n_N) \sim \text{Multinomial}(n, 1/N)$ seeds per current plant; each contributing plant selfed into n_k seeds

- *Reference:* Bernardo 2020 (page unverified)
- *Function:* bulk
- *Location:* R/select_schemes.R:156–169; .bulk_counts :470–472
- *Notes:* multinomial since the post-audit fix of F3 (was $n_{each} = \max(2, \lceil n/N \rceil)$ then n drawn without replacement)

V2-selection-16. Pedigree family size

$n_{each} = \max(1, \lceil \text{pop_size} / n_{sel} \rceil)$, trim at random

- *Reference:* Bernardo 2020 (page unverified)
- *Function:* pedigree
- *Location:* R/select_schemes.R:304–307

V2-selection-17. Tandem schedule

$t_g = \text{trait}[\lceil ((g - 1) \bmod L) + 1 \rceil]$

- *Reference:* DECISION-028
- *Function:* pedigree, recurrent_selection
- *Location:* R/select_schemes.R:291, 398

V2-selection-18. Intermating

each cross: pair ~ sample.int(np, 2) (no self), with replacement across crosses

- *Reference:* —
- *Function:* .intermate
- *Location:* R/select_schemes.R:542–545

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-selection-C1. General index response (context, not implemented)

$$\Delta H = i \text{Cov}(H, I) / \sigma_I$$

- *Variables:* $H = a' A$ aggregate breeding value; I selection criterion; i intensity
- *Reference:* Smith 1936 *Ann. Eugen.* 7:240–250 (Codex asserts pp. 241–242 Eq. 4; not re-verified here → page unverified); Hazel 1943 *Genetics* 28:476–490 (page unverified)
- *Function:* roxygen only
- *Location:* R/select_ind.R:16–21
- *Notes:* The general form; reduces to $ih^2\sigma_P$ (and to $G = \text{Cov}(A)$ in Smith–Hazel) only when $\text{Cov}(A, P) = V_A$, i.e. $\text{Cov}(A, D) = 0$ under HWE. Source of F4/O4.

V2-selection-C2. Singular Smith–Hazel extension

$$b = P^+ G a, P^+ = \text{Moore–Penrose via SVD, rank tol} = \max(\dim P) \varepsilon d_1$$

- *Variables:* P^+ pseudo-inverse; d_1 largest singular value
- *Reference:* Package numerical extension; no genetic primary source
- *Function:* .index_weights
- *Location:* R/select_ind.R:799–830
- *Notes:* Minimum-norm solution; warns on rank deficiency; exact inverse when full rank.

V2-selection-C3. Selfed / DH family BV correlation

$$r_{S1} = \frac{2(1+F)}{3+F}, r_{DH} = \frac{1+F}{2}$$

- *Variables:* F parental inbreeding; $r = A_{ij} / \sqrt{A_{ii}A_{jj}}$
- *Reference:* Package derivation from the tabular A (reconciler check: S1 sibs $A_{ij} = A_{PP} = 1 + F$, $A_{ii} = 1 + (1+F)/2$; DH $A_{ij} = 1 + F$, $A_{ii} = 2$ – algebra consistent); no primary page → UNVERIFIABLE attribution
- *Function:* select_ind() docs (family_relationship)
- *Location:* R/select_ind.R:94–98
- *Notes:* Reduces to 2/3 and 1/2 at $F = 0$.

V2-selection-C4. Sequential culling

$$S_0 = \{1..N\}; S_t = \text{Top}_{\max(1, \text{round}(c_t | S_{t-1}|))}(z_t; S_{t-1})$$

- *Variables:* c_t kept fraction per trait; S_t survivors
- *Reference:* Hazel & Lush 1942 *J. Hered.* 33:393–399 (page unverified)
- *Function:* .select_culling
- *Location:* R/select_ind.R:1067–1074
- *Notes:* Trait order matters; kept_at records $|S_t|$. Fable folded this into its culling row.

V2-selection-C5. Population pooling

$C = [C_1 C_2 \dots]$, $T = [T_1 T_2 \dots]$ subject to one shared map; ids `make.unique(sep = "_")`; pedigrees unioned by key

- *Variables:* C, T cis/trans haplotype matrices
- *Reference:* Package orchestration; no primary source
- *Function:* `c.Population`
- *Location:* `R/select_schemes.R:25–68`
- *Notes:* Map equality via `identical(snp, chr) + all.equal(pos, cm)` (O7).

V2-selection-C6. Single seed descent

$$P_{g+1} = \{\text{self}(i, 1) : i \in P_g\}$$

- *Variables:* one selfed seed per line per generation
- *Reference:* Bernardo 2020 (secondary; page unverified)
- *Function:* `single_seed_descent`
- *Location:* `R/select_schemes.R:102–118`
- *Notes:* Line count preserved; no selection.

V2-selection-C7. Recurrent-selection cycle

$$P_{c+1} = \bigcup_{j=1}^{n_x} \text{cross}(u_j, v_j, n_p), (u_j, v_j) \sim \text{sample.int}(n_p, 2) \text{ (no selfing)}$$

- *Variables:* n_x crosses; n_p progeny per cross
- *Reference:* Bernardo 2020; Falconer & Mackay 1996 (secondary; page unverified)
- *Function:* `recurrent_selection, .intermate`
- *Location:* `R/select_schemes.R:387–410; 354–370`
- *Notes:* Fable’s “Intermating” row covers only the pair draw (361-364); this row adds the cycle.

V2-selection-C8. Scheme RNG threading

one `set.seed(seed)` per wrapper, then every `selfcross()/cross()/sample.int()` consumes the ambient stream (`seed = NULL`)

- *Variables:* seed scheme seed
- *Reference:* DECISION-006/015 (project policy; not an equation)
- *Function:* all four wrappers, `.self_each, .intermate`
- *Location:* `R/select_schemes.R:111, 155, 283, 384; 521; 544`
- *Notes:* Valid seeds reproduce (both auditors); `.Random.seed` not restored (F13); seed not validated (O9).

Genomic relationship, optimum contribution, cross usefulness, MAS, MABC

Source report: v2-ocs-usefulness-marker.md

Entries: 19

V2-ocs-usefulness-marker-1. Genomic relationship

$$G = \frac{ZZ'}{2 \sum_j p_j(1 - p_j)}, \quad Z = M - 2p$$

- *Variables:* M ind×marker gene content 0/1/2; p_j allele freq (sample or base_freq)
- *Reference:* VanRaden 2008 *J Dairy Sci* 91:4414–4423, method 1 (paper verified; page unverified)
- *Function:* `g_matrix`
- *Location:* `R/select_ocs.R:71–92`
- *Notes:* monomorphic $p_j \in \{0, 1\}$ dropped :83–89; ridge blend :93–95

V2-ocs-usefulness-marker-2. Genomic inbreeding

$$F_i = G_{ii} - 1$$

- *Reference:* VanRaden 2008 (page unverified)
- *Function:* `g_matrix` (doc)
- *Location:* `R/select_ocs.R:16, 42–43`
- *Notes:* relative to the base used

V2-ocs-usefulness-marker-3. OCS objective

$$\max_{c \geq 0, 1'c=1} \quad c'g - \frac{\lambda}{2} c'Gc$$

- *Variables:* c contributions; g merit; G from `g_matrix`
- *Reference:* Meuwissen 1997 *J Anim Sci* 75:934–940 (verified; page unverified)
- *Function:* `optimum_contribution, .frank_wolfe`
- *Location:* `R/select_ocs.R:105–108, 601–624`
- *Notes:* `direction="low"` negates g :264

V2-ocs-usefulness-marker-4. Group coancestry

$$\bar{f} = \frac{1}{2} c'Gc$$

- *Reference:* Meuwissen 1997 (page unverified)
- *Function:* `optimum_contribution, .tune_lambda`
- *Location:* `R/select_ocs.R:319, 666`

V2-ocs-usefulness-marker-5. FW gradient / duality gap

$$\nabla = g - \lambda Gc; \quad \text{gap} = \max_k \nabla_k - \nabla'c$$

- *Reference:* Lacoste-Julien & Jaggi 2015 NIPS 28 (verified)

- *Function*: `.frank_wolfe`
- *Location*: R/select_ocs.R:601–607
- *Notes*: tol on gap :604

V2-ocs-usefulness-marker-6. Away-step and line search

$$d_{fw} = e_s - c, d_{aw} = c - e_a, \gamma^* = \min(\gamma_{max}, \frac{\nabla' d}{\lambda d' G d}), \gamma_{max}^{aw} = \frac{c_a}{1-c_a}$$

- *Reference*: Lacoste-Julien & Jaggi 2015 (verified)
- *Function*: `.frank_wolfe`
- *Location*: R/select_ocs.R:609–625
- *Notes*: exact maximiser of concave quadratic along d

V2-ocs-usefulness-marker-7. Penalty tuning

$$\text{bisection on } \lambda \text{ for } \frac{1}{2} c^*(\lambda)' G c^*(\lambda) = \bar{f}_{target}$$

- *Reference*: (monotone in λ : exchange argument)
- *Function*: `.tune_lambda`
- *Location*: R/select_ocs.R:652–718
- *Notes*: doubling to bracket :697–701

V2-ocs-usefulness-marker-8. Parent sampling

default method = "allocate": $n_i = \lfloor nc_i \rfloor$ plus one slot each for the $n - \sum_i \lfloor nc_i \rfloor$ largest remainders (slot order shuffled); method = "multinomial": $n_i \sim \text{Multinomial}(n, c)$

- *Reference*: package definition (documented)
- *Function*: `sample_parents`
- *Location*: R/select_ocs.R:429–441; `.allocate_slots` :473–494
- *Notes*: allocation is the default since the post-audit fix of OCS-F2 (owner decision D5, docs/DECISIONS.md); multinomial only on request

V2-ocs-usefulness-marker-9. Usefulness

$$U = \mu + i \sigma \text{ (or } \mu - i \sigma \text{ for "low")}$$

- *Variables*: μ, σ realized family mean/sd of additive value
- *Reference*: Zhong & Jannink 2007 *Genetics* 177:567–576; Lehermeier et al. 2017 *Genetics* 207:1651–1661 (verified; pages unverified)
- *Function*: `cross_usefulness`
- *Location*: R/select_usefulness.R:142–144
- *Notes*: family from crossing engine :140, :297–315

V2-ocs-usefulness-marker-10. Selection intensity

$$i(p) = \frac{\varphi(\Phi^{-1}(1-p))}{p}$$

- *Variables:* $p = \text{select_top}$
- *Reference:* standard normal-theory (Falconer & Mackay 1996, author-year)
- *Function:* `.intensity_from_p`
- *Location:* R/select_usefulness.R:161–163
- *Notes:* infinite-population value (1.755 at $p=0.1$)

V2-ocs-usefulness-marker-11. Fixed additive score

$$g_i = \sum_j x_{ij} e_j, \quad e_j = e_j^{raw} \sqrt{\pi_t} / \text{sd}(\text{comp})$$

- *Variables:* $x \in \{-1, 0, 1\}$; π_t layer prop; sd over template
- *Reference:* package definition (mirrors `.genetic_matrix`)
- *Function:* `.additive_model`, `.additive_gv`
- *Location:* R/select_usefulness.R:204–209, 228
- *Notes:* orthogonal layer uses α :194–198

V2-ocs-usefulness-marker-12. Average effect

$$\alpha = a + d(1 - 2p) = a + d(q - p)$$

- *Reference:* Falconer & Mackay 1996 (author-year)
- *Function:* `.avg_effect`
- *Location:* R/grammar_realize.R:182–184
- *Notes:* used by `.additive_model`:197, `marker_select` docs

V2-ocs-usefulness-marker-13. MAS feasibility

$$z_{mi} = x_{mi} \cdot f_m; \quad \text{met}_{mi} = [z_{mi} \geq \tau_m], \quad \tau = 0 \text{ carrier, } 1 \text{ homozygote; feasible} \iff \sum_m \text{met}_{mi} \geq k_{min}$$

- *Variables:* $f_m \in \{-1, +1\}$ favourable homozygote
- *Reference:* DECISION-029 (package definition)
- *Function:* `marker_select`
- *Location:* R/select_marker.R:116–119
- *Notes:* ranking :167

V2-ocs-usefulness-marker-14. MARS/oracle index

$$I_i = \sum_j w_j x_{ij}$$

- *Variables:* $w_j = a_j$ or α_j
- *Reference:* Lande & Thompson 1990 *Genetics* 124:743–756 (verified; page unverified)
- *Function:* `additive_value` (documented in `marker_select`)
- *Location:* R/cross_population.R:735–747; doc R/select_marker.R:20–33

V2-ocs-usefulness-marker-15. Recurrent-allele score

$$s_{mi} = \frac{x_{mi}r_m + 1}{2}$$

- *Variables:* $r_m = \pm 1$ recurrent homozygote dosage
- *Reference:* package definition
- *Function:* `.mabc_founders`
- *Location:* `R/select_mabc.R:311`
- *Notes:* informative iff $|r_m| = 1$, $d_m = -r_m$:309

V2-ocs-usefulness-marker-16. Recovery

$$R_i = \frac{\sum_{m \in B} w_m s_{mi}}{\sum_{m \in B} w_m}$$

- *Variables:* B background set; w equal / interval / user
- *Reference:* package definition (marker-observed, not IBD)
- *Function:* `.mabc_recovery`
- *Location:* `R/select_mabc.R:464`

V2-ocs-usefulness-marker-17. Expected recovery

$$R_t = 1 - 2^{-(t+1)}, R_t = \frac{1}{2} + \frac{1}{2}R_{t-1}$$

- *Variables:* t backcrosses after F1
- *Reference:* Mendelian; attributed to Frisch & Melchinger 2005 *Genetics* 170:909–917 (paper verified; location unverified)
- *Function:* doc only
- *Location:* `R/select_mabc.R:25–31, 227–229`
- *Notes:* MC: 0.756/0.870/0.934

V2-ocs-usefulness-marker-18. Interval weights

$$w_m = \left[\frac{p_m + p_{m+1}}{2} - \frac{p_{m-1} + p_m}{2} \right] - \text{overlap with excluded intervals, end cells to chromosome ends}$$

- *Variables:* p cM positions
- *Reference:* package definition
- *Function:* `.mabc_weights`
- *Location:* `R/select_mabc.R:520–529`
- *Notes:* per-chromosome sum = mapped – excluded length

V2-ocs-usefulness-marker-19. Foreground / recombinant / background staging

lexicographic: feasible $\rightarrow$ #flanks recurrent-homozygous $\rightarrow R_i \rightarrow$ seeded tie

- *Reference:* Frisch & Melchinger 2001 *Crop Sci* 41:1485–1494 (verified; page unverified)
- *Function:* `mabc_select`
- *Location:* `R/select_mabc.R:170–177, 194`

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-ocs-usefulness-marker-C1. Ridge blend of G

$$G^* = (1 - r) G + r I$$

- *Variables:* $r = \text{ridge} \in [0, 1]$
- *Reference:* package extension (no primary source claimed)
- *Function:* `g_matrix`
- *Location:* R/select_ocs.R:25–26, 44–50, 93–95
- *Notes:* Identity regularisation for a PD matrix; not VanRaden's pedigree- A blend. Fable mentioned it only in a Notes cell.

V2-ocs-usefulness-marker-C2. MABC foreground feasibility

carrier: $s_{ji} < 1 \forall j \in T$; donor homozygote: $s_{ji} = 0 \forall j \in T$, with $s_{ji} = (x_{ji}r_j + 1)/2$

- *Variables:* T target markers; $r_j = \pm 1$ recurrent homozygote code
- *Reference:* Frisch & Melchinger 2001 *Crop Sci* 41:1485-1494 (foreground stage; p. 1485 per Codex, page unverified by reconciler); rule itself is package/DECISION-029-style definition
- *Function:* `mabc_select`
- *Location:* R/select_mabc.R:168–174
- *Notes:* Under the documented shared-allele-coding precondition. Fable covered this in finding MABC-P1 but not as a map row.

Prediction: pedigree A matrix, BLUP/GBLUP, combining ability, progeny testing

Source report: v2-prediction.md

Entries: 19

V2-prediction-1. Tabular relationship (off-diagonal)

$$A_{kj} = \frac{1}{2} (A_{m(k),j} + A_{f(k),j}), j < k$$

- *Variables:* $m(k)$, $f(k)$ parents of k (processed parents-first)
- *Reference:* Emik & Terrill 1949 *J. Hered.* 40:51–55 (page unverified); Wright 1922 *Am. Nat.* 56:330–338 (page unverified)
- *Function:* `a_matrix`
- *Location:* R/select_blup.R:73–79
- *Notes:* unknown parent contributes 0; DH: $m = f$ so row = $A_{\{P, \cdot\}}$

V2-prediction-2. Inbreeding / diagonal

$$A_{kk} = 1 + F_k, F_k = \frac{1}{2} A_{m(k)f(k)}; \text{founder } A_{kk} = 1; \text{DH } A_{kk} = 2$$

- *Variables:* —
- *Reference:* as above

- *Function:* a_matrix
- *Location:* R/select_blup.R:80–88
- *Notes:* selfing: $F_t = 1 - (1/2)^t$ from a non-inbred founder (verified t = 1,2,3); founders forced F = 0 (PRED-F1)

V2-prediction-3. Variance ratio

$$\lambda = \sigma_e^2 / \sigma_A^2; \text{ with } h^2: \sigma_A^2 = h^2 \sigma_P^2, \sigma_e^2 = (1 - h^2) \sigma_P^2, \sigma_P^2 = \text{var}(\text{ref or pheno})$$

- *Variables:* —
- *Reference:* Henderson 1975 *Biometrics* 31:423–447 (page unverified)
- *Function:* predict_ebv, .blup_variances
- *Location:* R/select_blup.R:365, :676–681
- *Notes:* not inverted (verified)

V2-prediction-4. GLS mean

$$\hat{\mu} = (\mathbf{1}' V^{-1} \mathbf{1})^{-1} \mathbf{1}' V^{-1} y, V = K_{rr} + \lambda I$$

- *Variables:* r = phenotyped rows; V on the σ^2_A scale
- *Reference:* Henderson 1975 (page unverified)
- *Function:* predict_ebv
- *Location:* R/select_blup.R:376–386
- *Notes:* Cholesky of V is the PD check

V2-prediction-5. BLUP of u

$$\hat{u} = K_{.r}^{-1} V^{-1} (y - \mathbf{1} \hat{\mu}) \equiv \text{MME} \begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + \lambda K^{-1} \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

- *Variables:* Z selects one record per individual
- *Reference:* Henderson 1975 (page unverified)
- *Function:* predict_ebv
- *Location:* R/select_blup.R:387–388
- *Notes:* no K^{-1} needed; equals MME to 1e-10 (executed) and RR-BLUP for singular G

V2-prediction-6. PEV / reliability

$$\text{PEV}_i / \sigma_A^2 = K_{ii} - K_{ir} P K_{ri}, P = V^{-1} - V^{-1} \mathbf{1} (\mathbf{1}' V^{-1} \mathbf{1})^{-1} \mathbf{1}' V^{-1}; r_i^2 = 1 - \text{PEV}_i / (K_{ii} \sigma_A^2)$$

- *Variables:* $K_{ii} = 1 + F_i$
- *Reference:* Henderson 1975 (page unverified)
- *Function:* predict_ebv
- *Location:* R/select_blup.R:395–398
- *Notes:* $K = I: r^2 = h^2(1 - 1/n)$ (test line 24); equals $1 - C^{\{uu\}}_{ii} \lambda / K_{ii}$

V2-prediction-7. GBLUP marker back-solve

$$\hat{u}_m = Z'_r \alpha / [2 \sum_j p_j (1 - p_j)], \alpha = V^{-1}(y - \mathbf{1}\hat{\mu}); Z\hat{u}_m = \widehat{\text{GEBV}}$$

- *Variables:* $Z = M - 2p$ (VanRaden method 1)
- *Reference:* VanRaden 2008 *J. Dairy Sci.* 91:4414–4423 (page unverified)
- *Function:* `.gblup_marker_effects`
- *Location:* `R/select_blup.R:695–703`
- *Notes:* equals RR-BLUP (executed)

V2-prediction-8. Accuracy / bias

$$\text{acc} = \text{cor}(\hat{u}, u); \text{slope} = \text{Cov}(u, \hat{u}) / \text{Var}(\hat{u})$$

- *Variables:* —
- *Reference:* package (BLUP property $E[u | \hat{u}] = \hat{u}$)
- *Function:* `prediction_accuracy`
- *Location:* `R/select_blup.R:752–761`
- *Notes:* —

V2-prediction-9. Expected cross mean (one locus)

$$E[G_{ik}] = g_i g_k a + [g_i(1 - g_k) + (1 - g_i)g_k]d - (1 - g_i)(1 - g_k)a = -2d g_i g_k + (g_i + g_k)(a + d) - a, \\ g = x/2$$

- *Variables:* $x \in \{0, 1, 2\}$ gene content; genotypic values $-a, d, +a$
- *Reference:* package derivation from Falconer & Mackay 1996 genotypic values (page unverified)
- *Function:* `.expected_cross_means`
- *Location:* `R/select_combining.R:244–255`
- *Notes:* summed over loci; linearity $\Rightarrow$ linkage-free

V2-prediction-10. Tester-referenced average effect

$$\alpha_T = a + d(1 - 2p_T) = a + d(q_T - p_T); \text{GCA}_i \propto \frac{1}{2} \sum_j \alpha_{T,j} (x_{ij} - 2p_{C,j})$$

- *Variables:* p_T tester gamete freq
- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* `combining_ability` (doc)
- *Location:* `R/select_combining.R:30–37`
- *Notes:* = 1/2 DECISION-019 BV when $p_T = p$ (executed exact)

V2-prediction-11. Topcross / factorial GCA, SCA

$$g_i = \bar{Y}_{.i} - \mu, h_k = \bar{Y}_{.k} - \mu, s_{ik} = Y_{ik} - \mu - g_i - h_k$$

- *Variables:* μ grand mean of Y

- *Reference:* Sprague & Tatum 1942 *J. Am. Soc. Agron.* 34:923–932 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* `R/select_combining.R:321–324`
- *Notes:* balanced two-way LS

V2-prediction-12. Diallel GCA (method 4)

$$g_i = (m_i - \mu) \frac{p-1}{p-2} = \frac{pY_{i.} - 2Y_{..}}{p(p-2)}$$

- *Variables:* m_i mean of i 's $p-1$ crosses; μ mean of $p(p-1)/2$ crosses
- *Reference:* Griffing 1956 *Aust. J. Biol. Sci.* 9:463–493 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* `R/select_combining.R:312–316`
- *Notes:* algebraically identical to Griffing method 4 (executed)

V2-prediction-13. Diallel SCA (method 4)

$$s_{ij} = Y_{ij} - \mu - g_i - g_j = Y_{ij} - \frac{Y_{i.} + Y_{j.}}{p-2} + \frac{2Y_{..}}{(p-1)(p-2)}$$

- *Variables:* —
- *Reference:* Griffing 1956 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* `R/select_combining.R:317–318`
- *Notes:* $\sum_j s_{ij} = 0$ (executed)

V2-prediction-14. Broad-sense residual for simulated crosses

$$\sigma_e^2 = \text{Var}(G_{\text{ref}}) (1 - h^2) / h^2$$

- *Variables:* $G = A + D$ of progeny (default ref)
- *Reference:* Falconer & Mackay 1996 H^2 (page unverified)
- *Function:* phenotype_value via `.simulate_cross_means`
- *Location:* `R/cross_population.R:1061`; `R/select_combining.R:290–293`
- *Notes:* —

V2-prediction-15. Realized-scale template effects

$$a_j^* = a_j \sqrt{\pi_\ell} / \text{sd}(\text{raw component}_\ell) \text{ (same for d)}$$

- *Variables:* π_ℓ layer prop
- *Reference:* package (SPEC §2)
- *Function:* `.layer_scaled_effects ← template_effects`
- *Location:* `R/grammar_realize.R:364`; `R/select_combining.R:400–406`

- *Notes:* reproduces `genetic_values()` up to a constant (executed)

V2-prediction-16. Progeny-test expected mean

$$E[\bar{y}_i] = \frac{1}{2} \sum_j \alpha_{M,j} (x_{ij} - 2p_{M,j}) + c, \alpha_M = a + d(q_M - p_M)$$

- *Variables:* `p_M` mates' frequency
- *Reference:* package derivation; Falconer & Mackay 1996 α (page unverified)
- *Function:* `progeny_test` (doc)
- *Location:* `R/select_progeny.R:17–35`
- *Notes:* slope 0.503 on 200 progeny (executed)

V2-prediction-17. Progeny-test accuracy

$$r = \frac{V_A/2}{\sqrt{V_A V_P [1 + (n-1)h^2/4]/n}} = \sqrt{\frac{nh^2}{4 + (n-1)h^2}}; \text{crossover } n > \frac{4-h^2}{1-h^2}$$

- *Variables:* `n` half-sib records
- *Reference:* package derivation (classical; no page asserted)
- *Function:* `progeny_test` (doc)
- *Location:* `R/select_progeny.R:37–53`
- *Notes:* validated by simulation in `test-progeny.R:52–70`

V2-prediction-18. Mate sampling

mates for parent `j`: `sample.int` without replacement from distinct mates $\neq j$

- *Variables:* —
- *Reference:* package
- *Function:* `progeny_test`
- *Location:* `R/select_progeny.R:121–135`
- *Notes:* half-sib by construction

V2-prediction-19. Family mean

$$\bar{y}_i = \text{mean}\{y_k : \text{mother}(k) = i\}$$

- *Variables:* —
- *Reference:* package
- *Function:* `progeny_test`
- *Location:* `R/select_progeny.R:149–150`
- *Notes:* —

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-prediction-C1. Known-variance mixed model (the model BLUP solves)

$$y = \mathbf{1}\mu + Zu + e, \quad u \sim N(0, K\sigma_A^2), \quad e \sim N(0, I\sigma_e^2)$$

- *Variables:* K : A or G ; Z : one-record-per-individual incidence; μ : single fixed mean
- *Reference:* Henderson 1975 *Biometrics* 31:423–447 (page unverified)
- *Function:* `predict_ebv`
- *Location:* `R/select_blup.R`:174–181 (roxygen), :300–321 (inputs)
- *Notes:* Observable phenotypes only; intercept-only fixed effects; variance components known, no REML (DECISION-030). Fable’s map starts at the GLS solution and never states the model.

V2-prediction-C2. Supplied-covariance validation (correlation-scale PSD test)

$C_{ij} = K_{ij}/\sqrt{K_{ii}K_{jj}}$ over rows with $K_{ii} > 0$; accept iff $\lambda_{\min}(C) \geq -n\epsilon \max(1, |\lambda(C)|)$ and every zero-variance row is all-zero

- *Variables:* ϵ : `.Machine$double.eps`; n : number of eigenvalues
- *Reference:* Numerical-linear-algebra tolerance; no genetics source
- *Function:* `.check_relationship`
- *Location:* `R/select_blup.R`:587–650 (scaling :630–632, eigen :640, tolerance :644–645)
- *Notes:* PRED-01: when C is non-finite the code sets `ev <- -Inf`, which makes the tolerance ∞ and the test vacuous (CONFIRMED). Also $(K + t(K))/2$ at :265 can overflow for entries near `.Machine$double.xmax`.

V2-prediction-C3. VanRaden method-1 genomic relationship (consumed by GBLUP)

$$G = \frac{ZZ'}{2\sum_j p_j(1-p_j)}, \quad Z = M - 2p, \text{ markers with } p_j \in \{0, 1\} \text{ dropped; optional } G^* = (1 - \text{ridge})G + \text{ridge } I$$

- *Variables:* M : individuals $\times$ markers gene content 0/1/2; p : `base_freq` or current frequencies
- *Reference:* VanRaden 2008 *J. Dairy Sci.* 91:4414–4423 (page unverified)
- *Function:* `predict_ebv` $\rightarrow$ `g_matrix`
- *Location:* `R/select_blup.R`:337–343 (Codex wrote 168-173); `R/select_ocs.R`:44–98 (Codex wrote 44-92; core at :83, 90–92)
- *Notes:* Fable covered this only in the O1 checklist, not in its map. Inbred founders give $G_{ii} \approx 2$ while `a_matrix()` gives 1 (PRED-F1).

V2-prediction-C4. Progeny-test RNG draw order (reproducibility contract)

`set.seed(seed)  $\rightarrow$  for each parent  $j$ : sample.int of  $n$  distinct mates  $\neq j \rightarrow$  mate() meioses in plan order  $\rightarrow$  residual draws`

- *Variables:* one R stream; `seed = NULL` uses the ambient stream
- *Reference:* Package design, DECISION-012/027
- *Function:* `progeny_test`
- *Location:* `R/select_progeny.R`:113–114 (seed), :132 (mates), :138 (meioses), :144–145 (residual)
- *Notes:* Fable’s “Mate sampling” row covers only the first step. Ambient stream is not restored afterwards (PRED-F4).

Transcriptome: expression model, eQTL, mediation, NB counts, GREML mimic

Source report: v2-transcriptome.md

Entries: 25

V2-transcriptome-1. Expression model

$$E_{gi} = \mu_g + G_{gi} + R_{gi}$$

- *Variables:* μ : gene location (0 unless mimic); G genetic; R residual
- *Reference:* Falconer & Mackay 1996 (additive model, definition of h^2 ; page unverified)
- *Function:* `simulate_transcriptome`
- *Location:* R/transcriptome_simulate.R:616–618
- *Notes:* G, R drawn independently; mimic affine at :602-614

V2-transcriptome-2. Reference-centered dosage

$$Z_{ij} = x_{ij} - \bar{x}_j$$

- *Variables:* x in {-1,0,1}; marker mean on the reference panel
- *Reference:* own design (DECISION-020/021 fixed reference)
- *Function:* `simulate_transcriptome`
- *Location:* :308–310; predict :845–847
- *Notes:* mean, not 2p; equals x - 2p + 1 shift

V2-transcriptome-3. cis score

$$c_g = \sum_{j \in W(g)} \beta_{gj} Z_j, \beta \sim N(0, 1), k = 1 + \text{Bin}(2, 0.25)$$

- *Variables:* $W(g)$: same chr, $|pos - tss| \leq \text{cis_window}$, $\text{MAF} \geq 0.05$
- *Reference:* GTEx 2020 Science 369:1318-1330 (multiple cis-eQTL; verified)
- *Function:* `simulate_transcriptome`
- *Location:* :501–511
- *Notes:* window in physical bp

V2-transcriptome-4. trans factor

$$f_q = \sum_{k \in H(q)} \gamma_{qk} Z_k, |H(q)| = 1 + \text{Bern}(0.5), \text{loading } 1$$

- *Variables:* hubs outside all module genes' cis windows
- *Reference:* Albert et al. 2018 eLife 7:e35471 (trans hotspots; verified)
- *Function:* `simulate_transcriptome`
- *Location:* :451–471, :512
- *Notes:* factor inert if no distant marker

V2-transcriptome-5. Standardization

$$\tilde{v} = (v - \bar{v}) / s_v, s_v \text{ with } n - 1$$

- *Variables*: any component
- *Reference*: –
- *Function*: z1
- *Location*: :492–496
- *Notes*: returns NULL if s < 1e-9

V2-transcriptome-6. Additive genetic score

$$G^0 = \sqrt{\omega} \tilde{c} + \sqrt{1 - \omega} \tilde{t}; G = \sqrt{h^2} G^0 / s_{G^0}$$

- *Variables*: omega cis fraction (marginal)
- *Reference*: own design; SPEC section 4
- *Function*: simulate_transcriptome
- *Location*: :548–578
- *Notes*: joint scaling; cancellation fallback :555-558

V2-transcriptome-7. Epistatic blend

$$\tilde{e} = \sum_p \beta_p (\widetilde{Z_{j_p} Z_{k_p}} - \overline{Z_j Z_k}); G^0 = \sqrt{1 - \epsilon} \tilde{G}_{ct}^0 + \sqrt{\epsilon} \tilde{e}$$

- *Variables*: epsilon epistatic fraction, 1-2 pairs
- *Reference*: own design (a x a centered product)
- *Function*: simulate_transcriptome
- *Location*: :522–572
- *Notes*: prod_mean stored for predict

V2-transcriptome-8. Residual

$$R^0 = \sqrt{\kappa} \tilde{u}_{q(g)} + \sqrt{1 - \kappa} \tilde{\epsilon}_g; R = \sqrt{1 - h^2} R^0 / s_{R^0}$$

- *Variables*: u_q ~ N(0,1) shared per module; kappa residual module fraction
- *Reference*: own design
- *Function*: simulate_transcriptome
- *Location*: :585–595
- *Notes*: co-expression independent of h2

V2-transcriptome-9. Realized heritability

$$h_{real}^2 = \text{Var}(G) / \text{Var}(E) = h^2 / (1 + 2\text{Cov}(G, R))$$

- *Variables*: gr_cov = 2 Cov(G,R)
- *Reference*: Falconer & Mackay 1996 (definition); V2
- *Function*: simulate_transcriptome, predict
- *Location*: :619–628; :943–948

- *Notes:* unbounded above (TX-F1); the bounded allocation $h2_allocated = \text{Var}(G)/(\text{Var}(G)+\text{Var}(R))$ is reported separately (:627, :949)

V2-transcriptome-10. Effective coefficients

$$s_c = \sqrt{h^2(1-\epsilon)\omega/(s_{G_{ct}}^0 s_{G^0})}, s_t, s_e = \sqrt{h^2\epsilon}/s_{G^0}; \text{cis effect} = s_c\beta_j/s_c^{raw}; \text{trans_scale} = s_t/s_t^{raw}$$

- *Variables:* unit-scale truth
- *Reference:* own design
- *Function:* simulate_transcriptome
- *Location:* :636–646, :666, :673, :684
- *Notes:* reconstruct G exactly (verified 1e-8)

V2-transcriptome-11. Genetic budget

$$\text{Var}(G) = v_{cis} + v_{trans} + v_{epi} + 2\text{Cov}(c, t) + 2\text{Cov}(c, e) + 2\text{Cov}(t, e)$$

- *Variables:* –
- *Reference:* cf. DECISION-020 add_dom_cov pattern
- *Function:* simulate_transcriptome
- *Location:* :651–658
- *Notes:* closes to 1e-9

V2-transcriptome-12. Realized cis fraction

$$\omega_{real} = v_{cis}/\text{Var}(G), \text{Var}(G) \text{ the realized genetic variance (all covariance terms of V2-transcriptome-11 included)}$$

- *Variables:* –
- *Reference:* SPEC section 3 (“marginal”)
- *Function:* simulate_transcriptome
- *Location:* :659–661; predict :950
- *Notes:* reported as cis_fraction_realized; no longer identically omega on the reference (TX-F2 addressed): it equals omega only when the epistatic and covariance shares are zero

V2-transcriptome-13. Cross-population genetic value

$$G_g^{new} = Z_c^{new}\beta_g + \text{trans_scale}_g Z_H^{new}\gamma + \sum_p e_p(Z_j Z_k - \overline{Z_j Z_k}); Z^{new} = x^{new} - \bar{x}^{ref}$$

- *Variables:* fixed reference means
- *Reference:* own design (fixed-scale principle)
- *Function:* predict.transcriptome_sim
- *Location:* :829–848, :860–886
- *Notes:* scaled by stored gene_scale :921-927

V2-transcriptome-14. Mimic affine

$$E = \mu_g + \frac{\sqrt{V_g}}{s_{G+R}} [(G - \bar{G}) + (R - \bar{R})]$$

- *Variables:* mu_g, V_g from user matrix
- *Reference:* own design
- *Function:* simulate_transcriptome
- *Location:* :602–616
- *Notes:* exact moments (verified 1e-10)

V2-transcriptome-15. GRM

$$K = \frac{ZZ'}{m} \Big/ \text{diag}$$

- *Variables:* Z reference-centered dosages
- *Reference:* VanRaden 2008 J. Dairy Sci. 91:4414-4423 method 1 up to a constant (page listed in THEORY_REVIEW as “verify”; not re-verified here)
- *Function:* .tx_grm
- *Location:* R/transcriptome_mimic.R:24–30
- *Notes:* ratio to VanRaden G = 0.502 (constant)

V2-transcriptome-16. REML objective

$$-\ell_R(\delta) = \frac{1}{2}[(n-1)\log \text{RSS}(\delta) + \sum_i \log(\xi_i + \delta) + \log \sum_i \omega_i^2 / (\xi_i + \delta)], \text{RSS} = \frac{\sum w_i \eta_i^2 - (\sum w_i \omega_i \eta_i)^2 / \sum w_i \omega_i^2}{\sum w_i \omega_i^2}$$

- *Variables:* K = U diag(xi) U'; eta = U'y; omega = U'1; delta = sigma_e²/sigma_g²
- *Reference:* Kang et al. 2008 Genetics 178(3):1709-1723 (verified); Patterson & Thompson 1971 Biometrika 58:545-554 (page unverified)
- *Function:* .gremL_h2
- *Location:* R/transcriptome_mimic.R:76–84
- *Notes:* matches rrBLUP 8e-6

V2-transcriptome-17. GREML heritability

$$h^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2) = 1 / (1 + \hat{\delta})$$

- *Variables:* mean diag(K) = 1
- *Reference:* Yang et al. 2010 Nat. Genet. 42:565-569 (page unverified)
- *Function:* .gremL_h2
- *Location:* R/transcriptome_mimic.R:94–97
- *Notes:* boundaries 0/1 by comparison at log delta = +/-20

V2-transcriptome-18. Factor count

$$Q = \#\{\lambda_i(\frac{1}{T}E'_sE_s) > (1 + \sqrt{n/T})^2\}, \text{ capped to } [1, \min(50, n-2)]$$

- *Variables:* E_s standardized genes x ind

- *Reference:* Marchenko & Pastur 1967 Mat. Sb. 72:507-536 (page unverified)
- *Function:* `.tx_estimate_factors`
- *Location:* R/transcriptome_mimic.R:111-119
- *Notes:* floor 1 on noise (verified)

V2-transcriptome-19. kappa proxy

$$\hat{\kappa} = \text{clamp}_{[0,1]} \left[\left(r_w - \sqrt{h^2(1-\omega)} \right) / \sqrt{1-h^2} \right], r_w = \sum_{i \leq Q} s_i / (T - Q), s_i = \ell_i - 1 \text{ from } \lambda_i = \ell_i(1 + y/(\ell_i - 1)), y = T/n; s_i = 0 \text{ if } \lambda_i \leq (1 + \sqrt{y})^2$$

- *Variables:* λ_i leading Q eigenvalues of the gene-gene correlation matrix; h^2 per-gene GREML estimates; ω the assumed cis fraction (default 0.25)
- *Reference:* own estimator; the spike inversion is the Baik-Ben Arous-Péché spiked-model relation, named in the roxygen without a page-level source
- *Function:* `.tx_estimate_kappa`
- *Location:* R/transcriptome_mimic.R:149-172 (roxygen :120-148); called at R/transcriptome_simulate.R:395
- *Notes:* subtracts the GREML-implied genetic trans share (TX-F3 addressed); with $h^2 = 0$ no correction is applied

V2-transcriptome-20. NB observation

$$\log \mu_{gi} = \log L_i + \alpha_g + \sigma_g z_{gi}; Y_{gi} \sim \text{NB}(\mu_{gi}, \phi_g), \text{Var} = \mu + \phi \mu^2, \text{size} = 1/\phi, \phi = 0 \Rightarrow \text{Poisson}$$

- *Variables:* z per-gene standardized latent E
- *Reference:* McCarthy, Chen & Smyth 2012 NAR 40:4288-4297; Robinson & Smyth 2008 Biostatistics 9:321-332 (pages unverified); cited in the @references of `observe_counts` (R/transcriptome_counts.R:42, man/observe_counts.Rd)
- *Function:* `observe_counts`
- *Location:* R/transcriptome_counts.R:75-80, :98-102
- *Notes:* moments verified within 1.3%

V2-transcriptome-21. Layer score

$$T_x = c \sum_g w_g z_g, w = \text{slope} / \max |\text{slope}|, z_g = (E_g - \bar{E}_g) / s_{E_g}, c = \sqrt{\text{prop}} / s_{\sum w z}$$

- *Variables:* causal genes sparse
- *Reference:* own design (SPEC section 3; mirrors `additive()` scaling)
- *Function:* `.tx_raw`, `.transcriptome_matrix`
- *Location:* R/grammar_realize.R:454-476, :414-440
- *Notes:* $\text{Var}(T_x) = \text{prop}$ exactly

V2-transcriptome-22. Mediation split

$$T_{x,g} = c \sum_g w_g (G_g - \bar{G}_g) / s_{E_g}, T_{x,e} = T_x - T_{x,g}; \text{shares } \text{Var}(T_{x,g}) / V_P, \text{Var}(T_{x,e}) / V_P, 2\text{Cov} / V_P$$

- *Variables:* same denominator $\text{sd}(E_g)$ so $z_{\text{total}} = z_{\text{gen}} + z_{\text{env}}$
- *Reference:* own design; MESC estimand analogue (benchmarks README)
- *Function:* `.tx_raw(which="genetic")`, `.mediation_budget`, `mediation_split`
- *Location:* R/grammar_realize.R:458–476, :730–761; R/io_write.R:1267–1270
- *Notes:* sums to realized share ($1e-8$)

V2-transcriptome-23. Realized H2

$$H^2 = \text{Var}(\text{Gen} + T_{x,g}) / V_P$$

- *Variables:* Gen marker layers
- *Reference:* V2
- *Function:* `.genetic_value_matrix`, `.realized_h2`
- *Location:* R/grammar_realize.R:394–400, :1078–1106
- *Notes:* real source: $\text{Tx}_g = 0$

V2-transcriptome-24. Per-gene share

$$\text{var_explained}_g = (c w_g)^2 \cdot 1 / V_P$$

- *Variables:* –
- *Reference:* mirrors `.qtn_var`
- *Function:* `.tx_qtn_var`
- *Location:* R/io_write.R:1328–1345
- *Notes:* marginal; sum 0.36 vs prop 0.4 (co-expression cov)

V2-transcriptome-25. Layer sub-seed

$b \leftarrow (257 b + u_k) \bmod 2147483629$ over the UTF-8 codes u_k of the layer type ($b_0 = 0$); $s = (1009 \text{ seed} + 7919 b + 104729 \text{ occ}) \bmod (2^{31} - 1)$

- *Variables:* –
- *Reference:* R2 rule
- *Function:* `.layer_seed`, `transcriptome`
- *Location:* R/grammar_simulate_phenotype.R:1079–1092; R/transcriptome_layer.R:257–276
- *Notes:* order invariance verified

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-transcriptome-C1. Single-component LMM (model statement)

$$y = \mathbf{1}\mu + u + e, u \sim N(0, \sigma_g^2 K), e \sim N(0, \sigma_e^2 I); h^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2) = 1 / (1 + \delta)$$

- *Variables:* K : GRM with mean diagonal 1; $\delta = \sigma_e^2 / \sigma_g^2$
- *Reference:* Kang et al. 2008 *Genetics* 178(3):1709-1723 (EMMA; page/eq unverified)
- *Function:* `.gremL_h2`
- *Location:* `R/transcriptome_mimic.R:32–53 (roxygen), fit :58–97`
- *Notes:* Fable's map has the profile objective and h^2 rows but not the model statement; no identifiability guard (O4)

V2-transcriptome-C2. Calibration metrics

$$\text{bias} = \overline{\hat{\theta} - \theta}, \text{RMSE} = \sqrt{(\hat{\theta} - \theta)^2}$$

- *Variables:* θ target, $\hat{\theta}$ realized (h2, cis fraction)
- *Reference:* descriptive statistics; no source needed
- *Function:* `benchmark 01`
- *Location:* `benchmarks/01_h2_calibration.R:61–62 (h2), :93–94 (cis)`
- *Notes:* Cis-panel RMSE was 4e-16 by construction under the old $v_{cis} / (v_{cis} + v_{trans})$ definition (TX-F2); the benchmark now scores `cis_fraction_realized = vcis / Var( $G$ ) (:83)`, not re-run here

V2-transcriptome-C3. Marginal eQTL scan

$$r_j = \frac{(D_j - \bar{D}_j)^\top (y - \bar{y})}{\|D_j - \bar{D}_j\| \|y - \bar{y}\|}; \text{rank by } |r_j|; \text{best true rank} = \min_{j \in \text{true}} \text{rank}_j$$

- *Variables:* D_j dosage of candidate j ; y gene expression
- *Reference:* Pearson correlation (monotone in $|t|$ for one predictor); page unverified
- *Function:* `scan_one`
- *Location:* `benchmarks/02_eqtl_recovery.R:61–80 (r at :75, best rank :80)`
- *Notes:* Printed chance comparator :112–113 is $K / (D + 1)$; exact is $1 - \binom{D}{K} / \binom{D+m}{K}$ (O7)

V2-transcriptome-C4. Co-expression rank test

one-sided Wilcoxon rank-sum: H_1 : within-module $|r|$ stochastically $>$ between-module $|r|$; flag if $p < 0.05$

- *Variables:* pairwise gene-gene correlations of a genotype-free transcriptome
- *Reference:* Wilcoxon 1945 *Biometrics Bull.* 1(6):80-83 (page unverified)
- *Function:* `benchmark 03`
- *Location:* `benchmarks/03_coexpression_fp_control.R:55–70`

- *Notes:* Explicitly framed as the *naive* inference; “FPR” is relative to the stated truth $h^2 = 0$, not to the Wilcoxon null (O6, INFO)

V2-transcriptome-C5. Benchmark-04 mediation target

claimed $m_g \approx p h_*^2$; exact $m_g = \text{Var}(c \sum_g w_g (G_g - \bar{G}_g) / s_{E_g}) / V_P$ (reduces to $p \sum w_g^2 h_g^2 / \sum w_g^2 \cdot (p / V_P\text{-adj})$ only for independent genes)

- *Variables:* $p = \text{prop}$; w_g slopes; s_{E_g} sd of expression
- *Reference:* none cited; algebra
- *Function:* benchmark 04
- *Location:* benchmarks/04_mediation_recovery.R:9–12, 89–90, 114–116
- *Notes:* Generator run: actual 0.2771, $p \bar{h}^2$ 0.2560, exact 0.2771 (G)

V2-transcriptome-C6. TWAS Pearson test

$$t = r \sqrt{(n-2)/(1-r^2)}, p = 2F_{t_{n-2}}(-|t|)$$

- *Variables:* r gene-phenotype correlation
- *Reference:* standard Pearson-correlation t test; page unverified
- *Function:* cor_p
- *Location:* benchmarks/05_twas_power.R:103–112 (t at :110, p at :111)
- *Notes:* Equation correct; single-realization “power” (O10)

V2-transcriptome-C7. Bonferroni threshold

$$\alpha_{\text{gene}} = 0.05/T$$

- *Variables:* T genes tested
- *Reference:* Bonferroni; page unverified
- *Function:* benchmark 05
- *Location:* benchmarks/05_twas_power.R:114, 146–147
- *Notes:* Threshold correct

Input/output and QC: MAF, LD r2, PLINK pruning and blocks

Source report: v2-io-formats.md

Entries: 14

V2-io-formats-1. Allele frequency / MAF

$$p = \frac{\sum_i d_i}{2 n_{\text{called}}}, \quad \text{MAF} = \min(p, 1-p)$$

- *Variables:* $d_i \in \{0, 1, 2\}$ dosage of the allele coded 2 (from code_as), n_{called} = non-NA calls
- *Reference:* PLINK 1.9 --maf semantics (docs; Purcell 2007 for the tool)

- *Function:* filter_geno
- *Location:* R/qc_filter_geno.R:290–297
- *Notes:* n_called == 0 -> MAF 0; verified 4/10 on 0 0 0 2 2 NA NA NA

V2-io-formats-2. Monomorphic / MAF filters

keep iff $\text{MAF} > 0$; $\text{MAF} \geq m_{\text{above}}$; $\text{MAF} \leq m_{\text{below}}$

- *Variables:* —
- *Reference:* roxygen @param (>=, <=)
- *Function:* filter_geno
- *Location:* :311, :314, :318
- *Notes:* boundaries executed exact (exp3 F1)

V2-io-formats-3. Heterozygosity

$n_{\text{het}} = \#\{i : d_i = 1\}$; include iff $n_{\text{het}} > 0$, remove iff $n_{\text{het}} = 0$

- *Variables:* —
- *Reference:* roxygen
- *Function:* filter_geno
- *Location:* :301, :322, :324
- *Notes:* count, not rate; no rate threshold exists

V2-io-formats-4. Composite (genotypic) r^2

$$r^2 = \frac{(n \sum xy - \sum x \sum y)^2}{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}$$

- *Variables:* x, y dosages of two markers over n individuals (integer sums)
- *Reference:* PLINK 1.9 docs --indep-pairwise (“correlations between genotype allele counts”); plink_ld.c ld_prune (cov12*cov12*dxx)
- *Function:* .ld_sweep
- *Location:* :556–565
- *Notes:* = squared Pearson correlation; NA case falls back to cor(use="pairwise.complete.obs") :402

V2-io-formats-5. Pruning decision

prune iff $r^2 > \theta(1 + \varepsilon)$, $\varepsilon = 2^{-44}$; drop i iff $\text{MAF}_i < (1 - \varepsilon)\text{MAF}_j$ else drop j

- *Variables:* θ user threshold
- *Reference:* plink_ld.c ld_prune (fetched; “remove marker with lower MAF”)
- *Function:* .ld_sweep
- *Location:* :532, :663, :665–668
- *Notes:* tie $\rightarrow$ later marker pruned (earlier kept); executed exp3 F7

V2-io-formats-6. Window / step

window of w markers (or all markers within $w \cdot 1000$ bp of the start), slide by s , per chromosome in position order

- *Variables:* w, s
- *Reference:* PLINK 1.9 docs
- *Function:* `.ld_prune`, `.ld_sweep`
- *Location:* :492–506, :605–617 (win_at), :737–782 (slide)
- *Notes:* step rounded to integer ≥ 1 ; window fractional accepted (exp3 F7)

V2-io-formats-7. Haplotypic r^2

$$r^2 = \frac{(f_{11} - f_{1.}f_{.1})^2}{f_{1.}f_{2.}f_{.1}f_{.2}} = \frac{D^2}{p_1q_1p_2q_2}$$

- *Variables:* f_{11} ML haplotype frequency (cubic root with max log-likelihood), marginals $f_{1.}, f_{.1}$
- *Reference:* Hill & Robertson 1968 TAG 38:226–231 (page of eq. not verified); PLINK em_phase_hethet port
- *Function:* `.plink_hap_rsq`, `.plink_em_hethet`, `.plink_cubic_roots`
- *Location:* :796–809 (r^2 at :808), :822–879, :902–943
- *Notes:* independent EM agrees to 7 digits (exp3 F8)

V2-io-formats-8. VIF

$\text{VIF}_k = [R^{-1}]_{kk}$; prune $\arg \max_k \text{VIF}_k$ while $> v$

- *Variables:* R correlation matrix of the window's surviving markers
- *Reference:* PLINK 1.9 docs `--indep`; Purcell 2007 (VIF pruner)
- *Function:* `.ld_sweep` (vif branch)
- *Location:* :683–736 (solve :697, :721; test :725)
- *Notes:* singular ($\text{rcond} < 10^{-14}$) handling :362, :435–438, :533–557 UNVERIFIABLE vs source

V2-io-formats-9. Gabriel block MAF floor / tag

keep markers with $\text{MAF} \geq 0.05(1 - \varepsilon)$; tag = $\arg \max$ MAF in block

- *Variables:* —
- *Reference:* Gabriel 2002; PLINK `--blocks` (Haploview)
- *Function:* `.gabriel_blocks`
- *Location:* R/qc_ld_methods.R:25, :40
- *Notes:* classification thresholds (D' CI percentiles :814–815, informative fraction 0.95 :906) in `.plink_blocks_classify` R/qc_filter_geno.R:978–1053 and `.plink_blocks_chrom` :901–1015

V2-io-formats-10. Major-allele flip (frequency)

flip iff $\#\{d = 2\} > \#\{d = 0\}$

- *Variables:* homozygote counts (hets cancel)

- *Reference*: package convention (v1)
- *Function*: compute_flip
- *Location*: R/io_detect_format.R:362–364
- *Notes*: tie $\rightarrow$ no flip (root of IO-F2)

V2-io-formats-11. Reference flip

flip iff allele₁ $\neq$ ref

- *Variables*: per-marker labels
- *Reference*: package convention
- *Function*: compute_flip
- *Location*: :359
- *Notes*: only hapmap/table may use it (R/io_format_conversion.R:214–219)

V2-io-formats-12. Numeric coding

—101: (0, het, 2) $\mapsto$ (+1, 0, −1) unflipped, (−1, 0, +1) flipped; 012: (2, 1, 0) / (0, 1, 2); Dom: non-het $\rightarrow$ minor; Left: het $\rightarrow$ minor; Right: het $\rightarrow$ major

- *Variables*: raw dosage, flip
- *Reference*: v1 create_phenotypes transforms
- *Function*: numericalize_core
- *Location*: src/rust/src/numeric.rs:76–76, :138–145, :156–178
- *Notes*: imputation :47–52, :72–76 before the model transform

V2-io-formats-13. HWE

—

- *Variables*: —
- *Reference*: —
- *Function*: —
- *Location*: —
- *Notes*: **not implemented anywhere** (no exact/ χ^2 test)

V2-io-formats-14. Missing-rate filter

—

- *Variables*: —
- *Reference*: —
- *Function*: —
- *Location*: —
- *Notes*: **not implemented** (--geno/--mind absent)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-io-formats-C1. Raw genotype parsing and coding dispatch

call $\rightarrow g \in \{0, 1, 2, \text{NA}\}$ (0 = hom allele-1, 1 = het, 2 = hom allele-2); then major/het/minor = (1, 0, -1) or (2, 1, 0) after optional flip

- *Variables:* g raw dosage; flip from compute_flip
- *Reference:* Package-defined convention; no external source
- *Function:* parse_hapmap_chars_to_raw(), .apply_coding()
- *Location:* R/io_detect_format.R:223–326; R/io_read_formats.R:18–194
- *Notes:* Codex wrote 195–276; function closes at :277. Contract is violated by the SNPRelate readers (IO-F2): they deliver raw 2 = hom allele-1.

V2-io-formats-C2. Exact two-locus ML cubic

$$z^3 + az^2 + bz + c = 0, a = \frac{1}{2}(f_{11} + f_{22} - f_{12} - f_{21} - 3h), b = \frac{1}{2}(f_{11}f_{22} + f_{12}f_{21} + h(f_{12} + f_{21} - f_{11} - f_{22} + h)), \\ c = -\frac{1}{2}h f_{11}f_{22}$$

- *Variables:* f_{ab} known-haplotype shares; h = double-het count / $2N$ (hhs); z coupling increment
- *Reference:* Gaunt, Rodríguez & Day 2007 *BMC Bioinformatics* 8:428 (page unverified; **not in @references**); PLINK 1.9 plink_ld.c em_phase_hethet (no page)
- *Function:* .plink_em_hethet()
- *Location:* R/qc_filter_geno.R:822–879
- *Notes:* Codex wrote 646–713; verified coefficients at :674–677. Feasible roots clipped to $[0, h]$ (:679–686) and compared by log-likelihood — exact root selection, not iterative EM.

V2-io-formats-C3. Two-locus haplotype log-likelihood

$$\ell = c_{DH} \log(f_{11}f_{22} + f_{12}f_{21}) + \sum_{ab} c_{ab} \log f_{ab}, \text{ with } f_{11}, f_{22} += z, f_{12}, f_{21} += h - z$$

- *Variables:* c_{ab} unambiguous haplotype counts; c_{DH} double-heterozygotes
- *Reference:* Gaunt et al. 2007 (page unverified); PLINK plink_ld.c
- *Function:* .plink_calc_lnlike()
- *Location:* R/qc_filter_geno.R:884–894
- *Notes:* Codex wrote 716–728. Zero-count terms skipped (:724–727).

V2-io-formats-C4. Cubic real roots

$$Q = (a^2 - 3b)/9, R = (2a^3 - 9ab + 27c)/54; \text{ if } R^2 < Q^3 \text{ three real roots } -2\sqrt{Q} \cos(\frac{\theta + 2k\pi}{3}) - a/3, \\ \theta = \arccos(R/Q^{3/2})$$

- *Variables:* monic-cubic coefficients a, b, c
- *Reference:* Classical trigonometric Cardano (no page applicable); PLINK 1.9 plink_common.c cubic_real_roots
- *Function:* .plink_cubic_roots()
- *Location:* R/qc_filter_geno.R:902–943

- *Notes:* Codex wrote 731–777. PLINK constants EPSILON, π literal, and operation order retained (: 738–749).

V2-io-formats-C5. D' likelihood surface and class codes

$f_{11}(q) = f_{1.}f_{.1} + q \cdot \text{denom}$, $q = 0..100$ (D' percentile), other cells from marginals; $\ell(q)$ as above; classify by thresholds recomb_highci=89, strong_highci=97, strong_lowci=72, strong_lowci_outer=71

- *Variables:* q percentile of D' ; $\text{denom} = D_{\max}/100$ after sign orientation
- *Reference:* Gabriel et al. 2002 *Science* 296:2225–2229 (method page unverified); PLINK `plink_ld.c haploview_blocks_classify`
- *Function:* `.plink_calc_lnlike_quantile()`, `.plink_blocks_classify()`
- *Location:* `R/qc_filter_geno.R:952–966, :978–1053`
- *Notes:* Codex wrote 780–887. Thresholds verified at :814–815; 0.95 mass rule at :874.

V2-io-formats-C6. Gabriel/Haploview block acceptance

informative fraction = $0.95 + \varepsilon_{ish}$; two-marker span threshold $1 + \lfloor 3 \cdot \text{frac} \rfloor$, three-marker $\lfloor 6 \cdot \text{frac} \rfloor$; strong-pair fraction of informative pairs > 0.95 ; greedy largest-span non-overlapping blocks

- *Variables:* strong / recombination pair counts; span in bp
- *Reference:* Gabriel et al. 2002 (method page unverified); PLINK `plink_ld.c haploview_blocks`
- *Function:* `.plink_blocks_chrom()`, `.gabriel_blocks()`
- *Location:* `R/qc_filter_geno.R:1066–1180; R/qc_ld_methods.R:23–46`
- *Notes:* Codex wrote 890–1014 / 15–37. `inform_frac` at :906–908. Tag = highest-MAF member is a package convenience (`qc_ld_methods.R:40`).

V2-io-formats-C7. Per-QTN marginal realized variance share

$$k^2 = \pi_t / s_{\text{raw}}^2; v_j = k^2 \text{Var}(c_j) / V_P$$

- *Variables:* π_t layer prop for trait t ; $c_j = e_j G_j$ (additive), $e_j I(G_j = 0)$ (dominance), $a_j G_j + d_j I(G_j = 0)$ (orthogonal); V_P realized $\text{var}(y)$
- *Reference:* Package convention, docs/SPEC.md §2; no external source
- *Function:* `.qtn_var()`, `qtn_table()`
- *Location:* `R/io_write.R:1288–1312; :1400–1403`
- *Notes:* Marginal shares; with LD among causal loci they need not sum to π_t (roxygen :188–191). NA for vqtl/epistasis (:196–198).

V2-io-formats-C8. Per-gene transcriptome marginal share

$$k = \sqrt{\pi_t} / s_{\text{comp}}; w_g = \text{slope}_g / \max_g |\text{slope}_g|; v_g = (k w_g)^2 \text{Var}(z_g) / V_P \text{ with } \text{Var}(z_g) \in \{0, 1\}$$

- *Variables:* z_g standardized expression; constant gene $\rightarrow 0$
- *Reference:* Package convention, docs/SPEC.md; no external source
- *Function:* `.tx_qtn_var()`
- *Location:* `R/io_write.R:1328–1345`

- *Notes:* Codex wrote 235–251; closes at :252. Co-expression covariance not attributed to single genes.

V2-io-formats-C9. PLINK fixture allele serialization

$$-1 \mapsto A_1 A_1, 0 \mapsto A_1 A_2, 1 \mapsto A_2 A_2$$

- *Variables:* package dosage; allele column labels
- *Reference:* Package fixture; PLINK TPED format (manual page not cited)
- *Function:* fixture generator
- *Location:* data-raw/plink_parity_fixtures.R:26–34
- *Notes:* Codex wrote 19–35 (includes comments). r^2 , VIF, MAF invariant to a consistent A1/A2 swap. NA dosage would serialize as NA NA (IO-F15).

Version 1 (frozen legacy create_phenotypes engine)

V1 legacy: create_phenotypes core, residual/h2 machinery, linkage architecture, user QTNs

Source report: v1-core-linkage.md

Entries: 19

V1-core-linkage-1. Additive genetic value

$$g_i^A = \sum_k a_k x_{ik}, x \in \{-1, 0, 1\}$$

- *Variables:* a_k effect of QTN k ; x_{ik} dosage code
- *Reference:* Fernandes & Lipka 2020 (additive model; page unverified); F&M genotypic value scale $-a, d, +a$
- *Function:* genetic_effect
- *Location:* legacy_genetic_effect.R:109–112
- *Notes:* Coding centred at the heterozygote; not Fisher’s average effect

V1-core-linkage-2. Dominance value

$$g_i^D = \sum_k d_k \mathbf{1}[x_{ik} = 0]$$

- *Variables:* d_k dominance effect
- *Reference:* idem (page unverified)
- *Function:* genetic_effect
- *Location:* :61–75
- *Notes:* Het indicator; if a QTN has no het, $V_{D,k} = 0$ silently

V1-core-linkage-3. Epistatic value

$$g_i^E = \sum_k e_k \prod_{m \in S_k} x_{im}$$

- *Variables:* S_k set of epi_interaction markers

- *Reference*: idem, A x A only (page unverified)
- *Function*: genetic_effect
- *Location*: :77–89
- *Notes*: Product of -1/0/1 codes (sign flips)

V1-core-linkage-4. Centering

$$g_i \leftarrow g_i - \bar{g}$$

- *Reference*: -
- *Function*: genetic_effect
- *Location*: :93
- *Notes*: Hence phenotype mean = mean

V1-core-linkage-5. Degree of dominance

$$d_k = \delta a_k$$

- *Variables*: δ = degree_of_dom
- *Reference*: F&M partial/complete/over-dominance (d/a)
- *Function*: check_in
- *Location*: legacy_check_in.R:274
- *Notes*: Only with same_add_dom_QTN

V1-core-linkage-6. Geometric series

$$a_k = a^k, k = 1..n; \text{ with major QTN: } (a_{big}, a^1, \dots, a^{n-1})$$

- *Reference*: Fernandes & Lipka 2020 (geometric series; page unverified)
- *Function*: check_in
- *Location*: :786–788, 796–798, 810–812, 823–825
- *Notes*: Also applied to full-length custom vectors when mixed (F3)

V1-core-linkage-7. Residual variance

$$\sigma_e^2 = \frac{V_G(1-h^2)}{h^2}, V_G = \frac{1}{n-1} \sum (g_i - \bar{g})^2$$

- *Variables*: h^2 requested; V_G realized sample variance
- *Reference*: $h^2 = V_G/V_P$, $V_P = V_G + V_E$ (F&M; page unverified)
- *Function*: phenotypes
- *Location*: legacy_Phenotypes.R:216, 650, 1065, 1511
- *Notes*: Per rep for vary_QTN; n-1 denominator; no clamp on h^2 (F8)

V1-core-linkage-8. Single-trait phenotype

$$y_{ij} = g_i + e_{ij} + \mu, e_{ij} \sim N(0, \sigma_e^2)$$

- *Variables:* j = rep
- *Reference:* —
- *Function:* phenotypes
- *Location:* :669–674, 1517–1522
- *Notes:* rnorm inversion; seed per rep (F2)

V1-core-linkage-9. Multi-trait residual

$$\Sigma_e = D^{1/2} R D^{1/2}, D = \text{diag}(\sigma_{e,t}^2), R = \text{cor_res}$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :238–245, 1080–1086
- *Notes:* mvtnorm::rmvnorm (eigen method); R unvalidated

V1-core-linkage-10. Null trait (h2=0)

$$y_{ij} \sim N(\mu, 1); \text{multi: } N(\mu, R)$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :533–534, 86–92
- *Notes:* Unit variance, not V_G -scaled

V1-core-linkage-11. Realized heritability

$$\hat{h}_{tj}^2 = V_{G,t} / \widehat{\text{Var}}(y_{tj}), \text{ averaged over } j$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :248–251, 336, 675–676, 876
- *Notes:* Diagnostic only

V1-core-linkage-12. Per-QTN PVE

$$\text{PVE}_{kj} = \text{Var}(a_k x_k) / \widehat{\text{Var}}(y_j)$$

- *Reference:* -
- *Function:* phenotypes
- *Location:* :260–268, 685–693
- *Notes:* Ignores LD covariance; rep-1 numerator under vary_QTN multi-trait (F12)

V1-core-linkage-13. Residual seed

$$s_j = \lfloor (s_0 + j) \cdot \text{round}(10h_1^2) \rfloor; \text{null: } s_0 + j$$

- *Variables:* s_0 master seed
- *Reference:* - (roxygen :161–167 states a different formula)
- *Function:* phenotypes
- *Location:* :234, 665, 1076, 1513; 82, 529, 924, 1374
- *Notes:* F2

V1-core-linkage-14. LD-QTN seed

$s = (s_0 \cdot a) + z$ (add), $+ \text{rep}$ (dom), $+ x$ (direct resample)

- *Variables:* a attempt, z rep, x QTN index
- *Reference:* -
- *Function:* qtn_linkage
- *Location:* legacy_QTN_linkage.R:110, 383, 645, 971, 1220, 1243, 1455, 1479
- *Notes:* Same seed re-set before each inner resample -> deterministic walk (F4)

V1-core-linkage-15. LD measure

$\ell = |\text{LD}_{\text{method}}(x_j, x_i)|$, accept iff $\ell_{\min} \leq \ell \leq \ell_{\max}$

- *Variables:* method in {composite, r, dprime, corr}
- *Reference:* SNPRelate snpgdsLDpair docs (primary source for the composite measure not cited in package: UNVERIFIABLE)
- *Function:* qtn_linkage
- *Location:* :150–151, 176–177, 1025–1028, 1270–1273
- *Notes:* Correlation scale, not r^2 (INFO-3); LD_between_QTNs unsigned-ness not applied (:198–199)

V1-core-linkage-16. Indirect selection

$\text{QTN}_1 = \text{first } i > j \text{ with } \ell(j, i) \leq \ell_{\max}$; $\text{QTN}_2 = \text{first } i' < j \text{ likewise}$; require both $\geq \ell_{\min}$

- *Variables:* j = intermediate (“cause”) marker
- *Reference:* Fernandes & Lipka 2020 “indirect” (page unverified)
- *Function:* qtn_linkage
- *Location:* :125–212
- *Notes:* Index-based walk, chromosome-agnostic; trait 1 <- higher index (sup)

V1-core-linkage-17. Direct selection

$\text{QTN}_2 = \text{whichever of } j \pm 1 \text{ has } \ell \text{ in window, nearest } \ell_{\max}$; else resample j

- *Reference:* idem “direct”
- *Function:* qtn_linkage
- *Location:* :1229–1308 (fixed), :978–1062, 1465–1546 (unfixed)
- *Notes:* Fixed branch adjacent-only (INFO-4); unfixed branches F5

V1-core-linkage-18. MAF

$$p = \frac{\sum_i x_i + n}{2n}, \text{MAF} = \min(p, 1 - p)$$

- *Variables:* n taxa
- *Reference:* standard allele frequency from dosage
- *Function:* constraint, qtn_linkage, qtn_from_user
- *Location:* legacy_constraint.R:47–50; QTN_linkage.R:314–317; qtn_from_user.R:237–240
- *Notes:* Strict bounds in constraint (F23)

V1-core-linkage-19. Genetic correlation (given cor)

$$G_s = \text{scale}(G); W = L_g^{-1} G_s^\top; T = L_c W \text{ with } L_g L_g^\top = \text{cov}(G_s), L_c L_c^\top = \text{cor}; T_t \leftarrow T_t s_t + \bar{g}_t$$

- *Reference:* Cholesky whitening/colouring (standard); PleioArch not used in v1
- *Function:* base_line_multi_traits (not in group)
- *Location:* legacy_Base_line_multi_traits.R:167–183
- *Notes:* Realized sample correlation == cor exactly (INFO-2); upper triangle only (F18)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V1-core-linkage-C1. Heterozygote eligibility indicator

$$I_{het}(x) = \mathbf{1}[x = 0]$$

- *Variables:* $x \in \{-1, 0, 1\}$; 0 = Aa under the documented coding
- *Reference:* Package coding contract (create_phenotypes.R:100–102, roxygen “aa = -1, Aa = 0 and AA = 1”); external page UNVERIFIABLE
- *Function:* constraint() (correct); create_phenotypes() guard (wrong code)
- *Location:* legacy_constraint.R:28–42; legacy_create_phenotypes.R:100–102, 848–935
- *Notes:* constraint() tests == 0; the guard tests b == 1 (V1C-F11 / Codex F06). Verified: hets <- NULL at :848, if (any(!unlist(hets))) at :922, block closes at :935.

V1-core-linkage-C2. Reported residual “sample correlation”

$$R_{reported} = \text{cov2cor}\left(\frac{1}{r} \sum_{z=1}^r \Sigma_{E,z}\right), \Sigma_{E,z} = D^{1/2} R D^{1/2}$$

- *Variables:* r replicates; $\Sigma_{E,z}$ the target covariance, not realized residuals
- *Reference:* Implementation-defined; the “sample” label conflicts with the paper’s stated diagnostic (page unverified)
- *Function:* phenotypes()
- *Location:* legacy_Phenotypes.R:483–491, 1335–1343 (verified: matrix(0,...) at :484/:1336, cov2cor at :488/:1340)
- *Notes:* Equals cor_res exactly whenever cor_res is a valid correlation matrix (V1C-F10 / Codex F10). Not an estimate.

V1-core-linkage-C3. QTN anchor sampling without replacement

$J \sim \text{Sample}(\mathcal{E}, m, \text{replace} = \text{FALSE})$ after `set.seed(seed*s+z)` (add) / `set.seed(seed*s+z+rep)` (dom)

- *Variables:* $\mathcal{E}$ = index (constraint-filtered rows or all rows); m = QTN count; s outer attempt, z rep
- *Reference:* Implementation-defined; Fernandes & Lipka 2020 (random QTN selection; page unverified)
- *Function:* `qtn_linkage()`
- *Location:* `legacy_QTN_linkage.R`:109–113, 382–386, 644–648, 970–974, 1219–1223, 1454–1458 (verified: `while (s <= 10 & border) at :109,382,644,970,1219,1454; sample(index, ...)` at :113,386,648,974,1223,1458)
- *Notes:* Uses R's `sample()` under `RNGversion("3.5.1")` ("Rounding"). Constraints apply to anchors only; walked partners are unfiltered (Codex F14 / Fable F23).

V1-core-linkage-C4. User marker resolution and order

$I = (\text{match}(m_1, S), \dots, \text{match}(m_k, S))$ realised as `genotypes[genotypes$snp %in% i, ]` then `a[i, ]` (named-row indexing in requested order)

- *Variables:* m requested IDs; S genotype IDs
- *Reference:* Implementation-defined API algorithm
- *Function:* `qtn_from_user()`
- *Location:* `legacy_qtn_from_user.R`:1358–1388 (add), :1447–1460 (dom), :1517–1530 (epi), :1589–1602 (var) (verified: `selected_snps <- genotypes$snp %in% i at :1361, 1450, 1520, 1592)`
- *Notes:* Only `sum(selected_snps) == 0` is rejected; a partially missing list yields an all-NA row named `Chr_NA_NA` (Codex F08; public outcome is a cryptic error, R5). Single-trait assembly wraps the result one level too deep (V1C-F1).

V1-core-linkage-C5. Epistatic interaction diagnostic expansion

$n_{\text{groups}} = n_{\text{markers}}/k$; `QTN = rep(1:n_groups, each = k)`; effect column `rep(effect, each = n_groups)` (should be `each = k`)

- *Variables:* k = `epi_interaction`
- *Reference:* Implementation-defined API grouping
- *Function:* `qtn_from_user()`
- *Location:* `legacy_qtn_from_user.R`:1582–1605 (verified: `e_len <- lengths(QTN_list$epi)/epi_interaction at :1532, each = epi_interaction at :1541, each = e_len[i] at :1544)`
- *Notes:* Correct only when $n_{\text{groups}} = k$ (V1C-F13 / Codex F09).

V1-core-linkage-C6. (reconciler-added) Direct-LD acceptance sentinel

Loop runs while $(\ell_{\text{sup}} \notin [\ell_{\text{min}}, \ell_{\text{max}}]) \wedge (\ell_{\text{inf}} \notin [\ell_{\text{min}}, \ell_{\text{max}}])$ with initial $\ell_{\text{sup}} = \ell_{\text{inf}} = 1$; on exit partner $= i - 1$ or $i_2 + 1$

- *Variables:* sentinel value 1
- *Reference:* -
- *Function:* `qtn_linkage()`
- *Location:* `legacy_QTN_linkage.R`:1269–1269 (fixed add branch); same pattern :123–125 (indirect add), :976–983, :1467–1473
- *Notes:* When $\ell_{\text{max}} = 1$ the sentinel is "in window", the loop never executes and the partner is the anchor itself (RX-1, R6e); indirect errors "object 'i' not found" (R6f).

V1 legacy: pleiotropic and partially pleiotropic QTNs, genetic effects, vQTL, baselines

Source report: v1-pleio-effects.md

Entries: 18

V1-pleio-effects-1. Additive genetic value

$$g_i^A = \sum_{k=1}^{n_A} x_{ik} a_k$$

- *Variables:* $x_{ik} \in \{-1, 0, 1\}$ (+1 = major-allele homozygote), a_k additive effect
- *Reference:* Fernandes & Lipka 2020, Implementation (page unverified); Falconer & Mackay 1996 ch. 7 (page unverified)
- *Function:* genetic_effect
- *Location:* R/legacy_genetic_effect.R:111–57
- *Notes:* Not Fisher’s average-effect decomposition; no allele-frequency scaling

V1-pleio-effects-2. Geometric effect series

$$a_k = a^k, k = 1, \dots, n; \text{ with big QTN } (a_{\text{big}}, a^1, \dots, a^{n-1})$$

- *Variables:* $a = \text{add_effect}$, $n = \text{QTN count}$
- *Reference:* Fernandes & Lipka 2020 (“geometric series of effect sizes”, page unverified)
- *Function:* check_in
- *Location:* R/legacy_check_in.R:905–837
- *Notes:* Length-mismatch recycling (F3); mixed-type conversion (F4); applies equally to dom_, epi_, var_effect

V1-pleio-effects-3. Dominance genetic value

$$g_i^D = \sum_{k=1}^{n_D} \mathbb{1}[x_{ik} = 0] d_k$$

- *Variables:* d_k dominance effect; same_add_dom_QTN: $d_k = a_k \cdot \text{degree_of_dom}$
- *Reference:* paper Table 1 (UNVERIFIABLE); code
- *Function:* genetic_effect
- *Location:* R/legacy_genetic_effect.R:120–77; R/legacy_check_in.R:274
- *Notes:* Heterozygote indicator only; hetless locus contributes 0 with var_dom=0

V1-pleio-effects-4. Epistatic genetic value

$$g_i^E = \sum_{m=1}^{n_E} e_m \prod_{l \in S_m} x_{il}, |S_m| = \text{epi_interaction}$$

- *Variables:* e_m interaction effect
- *Reference:* paper “additive x additive epistatic” (page unverified)
- *Function:* genetic_effect
- *Location:* R/legacy_genetic_effect.R:151–91

- *Notes:* **Uncentered** dosage product; any heterozygote zeroes the term; k -way allowed; epi_interaction=1 errors

V1-pleio-effects-5. Total genetic value

$$g_i = (g_i^A + g_i^D + g_i^E) - \bar{g}$$

- *Variables:* —
- *Reference:* —
- *Function:* genetic_effect
- *Location:* R/legacy_genetic_effect.R:186–96
- *Notes:* Mean-centred; mean added later

V1-pleio-effects-6. Component variances

$$V_A = \widehat{\text{Var}}(g^A) \text{ etc. (sample, } n - 1); \text{ per-QTN } \text{Var}(x_{.k}a_k)$$

- *Variables:* —
- *Reference:* —
- *Function:* genetic_effect
- *Location:* R/legacy_genetic_effect.R:115,59,69,76,87,90
- *Notes:* Per-QTN variances ignore LD covariance; $\Sigma \neq V_A$

V1-pleio-effects-7. MAF of a QTN

$$p = \frac{1}{2} \left(\frac{\sum_i x_i + n}{n} \right), \text{MAF} = \min(p, 1 - p)$$

- *Variables:* n = individuals
- *Reference:* —
- *Function:* qtn_pleiotropic, qtn_partially_pleiotropic, constraint
- *Location:* R/legacy_QTN_pleiotropic.R:205–165; ...partially...R:165–169; R/legacy_constraint.R:47–43
- *Notes:* Rounded to 4 dp in files; NA-unsafe

V1-pleio-effects-8. QTN sampling seeds (pleiotropic)

$$\text{add: } s + i; \text{ dom: } s + i + r; \text{ var: } 2s + i; \text{ epi: } 2s + i + r$$

- *Variables:* s = seed, i = rep index, r = rep (1 unless vary_QTN)
- *Reference:* DECISION-008 (legacy seed math)
- *Function:* qtn_pleiotropic
- *Location:* R/legacy_QTN_pleiotropic.R:174,318,408,485
- *Notes:* Collisions (F8); sets not disjoint

V1-pleio-effects-9. QTN sampling seeds (partially)

pleio add $s + j$; spec add $s + i + j$; pleio dom $s + j + r$; spec dom $s + i + j + r$; pleio epi $2s + j$; spec epi $2s + i + j$

- *Variables:* j = rep, i = trait
- *Reference:* —
- *Function:* qtn_partially_pleiotropic
- *Location:* ...partially...R:125,153,446,475,621,637
- *Notes:* Pools: setdiff (disjoint) except F6

V1-pleio-effects-10. Het re-sample loop

re-draw while $\neg \exists(i, k) : x_{ik} = 0$, ≤ 10 tries, same seed, pool \ rejected

- *Variables:* —
- *Reference:* —
- *Function:* both QTN samplers
- *Location:* R/legacy_QTN_pleiotropic.R:180–145,324–333; ...partially...R:131–140,160–170,452–461,481–493
- *Notes:* “any het in the whole set”, not per QTN

V1-pleio-effects-11. Whitening

$$G_s = \text{scale}(G), C_g = \text{cov}(G_s), W = G_s L_g^{-\top}, L_g L_g^{\top} = C_g$$

- *Variables:* $G = n \times t$ genetic values
- *Reference:* Fernandes & Lipka 2020, Implementation (whitening/colouring paragraph; page unverified); Kessy et al. 2018 (page unverified)
- *Function:* base_line_multi_traits
- *Location:* R/legacy_Base_line_multi_traits.R:151–109,178–184
- *Notes:* C_g passed through make_pd

V1-pleio-effects-12. Colouring & rescale

$$T = W L_c^{\top}, L_c L_c^{\top} = \text{cor}; \tilde{g}_{.i} = T_{.i} \text{sd}(G_{.i}) + \overline{G_{.i}}$$

- *Variables:* cor user matrix ($t \times t$ only)
- *Reference:* same
- *Function:* base_line_multi_traits
- *Location:* ...multi_traits.R:160–120,185–194
- *Notes:* Exact sample correlation; trait ≥ 2 effects/PVE not effective (F2); variance preserved iff $\text{diag}(\text{cor}) = 1$

V1-pleio-effects-13. PD repair

$$\tau = \max(0, 2n \varepsilon \max |\lambda| - \lambda), m' = \text{round}(m + V \text{diag}(\tau) V^{\top}, 2)$$

- *Variables:* λ, V eigen-pairs of m
- *Reference:* none (ad hoc)
- *Function:* make_pd
- *Location:* R/legacy_make_pd.R:61–21
- *Notes:* Not a correlation matrix afterwards; rounding re-breaks PD (F1)

V1-pleio-effects-14. Residual variance (context)

$$\sigma_e^2 = v_g/h^2 - v_g, v_g = \widehat{\text{Var}}(\tilde{g}); \text{multi-trait } \Sigma_e = \text{diag}(\sigma_e) \text{cor_res} \text{diag}(\sigma_e)$$

- *Variables:* —
- *Reference:* paper: trait variance = genetic + error (page unverified)
- *Function:* phenotypes
- *Location:* R/legacy_Phenotypes.R:216,239,650
- *Notes:* Post-transform v_g ; seed $(s + j) \cdot \text{round}(10h_1^2)$

V1-pleio-effects-15. vQTL residual scale

$$\sigma_i = 1 + \sum_k v_k(x_{ik} + 1); k = \sqrt{(1/h^2 - 1)}/\text{median}_i(\sigma_i); e_{ij} \sim N(0, (k\sigma_i)^2); y_{ij} = \text{scale}(g)_i + e_{ij} + \mu$$

- *Variables:* $v_k = \text{var_effect}, x + 1 \in \{0, 1, 2\}$
- *Reference:* Murphy et al. 2022 Heredity 129:93–102 (form UNVERIFIABLE)
- *Function:* vQTL
- *Location:* R/legacy_vQTL.R:89,62–71,76–89
- *Notes:* g standardised to unit variance; SD linear in dosage (not log-linear); h^2 exact at median σ ; no guard on $\sigma \leq 0$

V1-pleio-effects-16. vQTL sample heritability

$$\hat{H}^2 = \text{mean}_j 1/\widehat{\text{Var}}(y_{.j})$$

- *Variables:* —
- *Reference:* —
- *Function:* vQTL
- *Location:* R/legacy_vQTL.R:251–193
- *Notes:* Valid because $\text{Var}(g) = 1$; wrong for $\text{nrow}(h2) > 1$ (F10)

V1-pleio-effects-17. Numeric normalisation

if no -1 and some 2 : $x \leftarrow x - 1$; NA $\leftarrow \{\text{Middle } 0, \text{Minor } -1, \text{Major } 1, \text{else } 0\}$

- *Variables:* —
- *Reference:* —
- *Function:* genotypes/numeric_df

- *Location:* R/legacy_Genotypes.R:79–62
- *Notes:* 0/1-only data not shifted (F12); in-memory twin at create_phenotypes.R:610–459 differs

V1-pleio-effects-18. maf_cutoff filter

keep $\min(p, 1 - p) \geq \text{cutoff}$, $p = \text{mean}((x + 1)/2)$

- *Variables:* —
- *Reference:* —
- *Function:* genotypes
- *Location:* R/legacy_Genotypes.R:163–111
- *Notes:* Inclusive; constraints are strict

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V1-pleio-effects-C1. Fully pleiotropic QTN-set construction

$S_1 = \dots = S_T = S$, $S \sim \text{sample w/o replacement from index}$; epistasis draws $w m_E$ physical loci and groups every w consecutive loci

- *Variables:* S shared causal set, T traits, $w = \text{epi_interaction}$, $m_E = \text{epi_QTN_num}$
- *Reference:* Fernandes & Lipka 2020 *BMC Bioinformatics* 21:491 (architecture description; page unverified)
- *Function:* qtn_pleiotropic
- *Location:* R/legacy_QTN_pleiotropic.R:169–201 (add), 313–345 (dom), 403–423 (var), 480–500 (epi)
- *Notes:* Effect-type sets drawn independently (not disjoint, F8); A/D share only under same_add_dom_QTN, A/V under same_mv_QTN; het loop requires $\$ \geq \1 heterozygote anywhere in the set

V1-pleio-effects-C2. Partial-pleiotropy locus construction

$S_t = S_P \cup S_{T,t}$, intended $S_{T,t} \cap S_{T,u} = \emptyset$ ($t \neq u$); distinct-locus demand $|S_P| + \sum_t |S_{T,t}|$

- *Variables:* S_P shared, $S_{T,t}$ trait-specific (drawn from setdiff pools)
- *Reference:* Fernandes & Lipka 2020 (partial pleiotropy; page unverified)
- *Function:* qtn_partially_pleiotropic
- *Location:* R/legacy_QTN_partially_pleiotropic.R:165–233 (add+dom-loop), 294–360 (add only), 441–533 (dom), 616–684 (epi)
- *Notes:* Disjointness violated by the het re-sampling loop (F6/O8); preflight demands $\sum_t (|S_P| + |S_{T,t}|)$ instead (O9); a single zero $|S_{T,t}|$ crashes (O4)

V1-pleio-effects-C3. Epistatic physical-marker demand

$N = w m_E$; partial: $N = w (|S_{P,E}| + \sum_t |S_{T,E,t}|)$

- *Variables:* $w = \text{epi_interaction}$
- *Reference:* none (implied by implementation)
- *Function:* qtn_pleiotropic, qtn_partially_pleiotropic
- *Location:* R/legacy_QTN_pleiotropic.R:524–532; R/legacy_QTN_partially_pleiotropic.R:586–611 (also :654)
- *Notes:* Sampling uses the multiplier; the constraint preflight (pleiotropic.R:105–109, partially.R:137–137) does not (O9)

Appendix A (at the audit commit, historical) — automated verification of cited locations

Every file:line reference above was checked against the worktree at HEAD c511c6f. Status meanings: **ok** = file exists, range inside file, and the named function occurs within the range or the 60 lines above it; **fn-in-file-only** = the function is defined in that file but not near the cited lines (usually a helper called from the cited lines); **fn-not-found** = the named function string does not occur in the cited file; **missing-file / beyond-eof** = the reference is wrong.

272 of 395 cited ranges verified *ok*; the exceptions are listed below (they are informational: a helper cited by its caller, or a reference into a non-R file).

Group	Topic	Function	Cited location	Status
v2-grammar/codex	Additive MAF effect scaling	.pleio_draw	R/effects_pleioarch.R:159	fn-in-file-only
v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	effects_pleioarch.R:139-141	fn-in-file-only
v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	effects_pleioarch.R:536	fn-in-file-only
v2-effects-arch	Allele -> genotype scaling	.pleio_draw; .marker_maf_ref	effects_pleioarch.R:143-159	fn-in-file-only
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_simulate_phenotypes.R:452-455	fn-not-found
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_realize.R:370	fn-in-file-only
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_realize.R:408	fn-in-file-only
v2-effects-arch	Effective per-component target	.pleio_unit_effects	grammar_realize.R:510	fn-in-file-only
v2-effects-arch	Total-correlation target	.pleio_total_cor_chrom	grammar_realize.R:605	fn-not-found
v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:112-122	fn-in-file-only
v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:114-114	fn-in-file-only
v2-effects-arch	Indirect flanks	.draw_qtn_ld	arch_ld.R:123-153	fn-in-file-only
v2-crossing	Heterozygote phasing	as_population	R/cross_population.R:119	fn-in-file-only
v2-crossing	Dosage from strands	dosages	genome.rs:195-200	fn-not-found
v2-crossing	Cross / self / DH	mate_haplotypes	R/cross_mating.R:102	fn-in-file-only
v2-crossing	Synthetic map rate	synthetic_map	R/cross_map.R:157	fn-in-file-only
v2-crossing	Synthetic map integration	synthetic_map	R/cross_map.R:161	fn-in-file-only
v2-crossing	Retention fractions (doc only)	roxygen	R/cross_breed.R:68	fn-not-found
v2-crossing	Pedigree key	.mating_pedigree, .stable_key	hash.rs:10-21	fn-not-found
v2-rust-core	Crossover count (count-location)	.draw_meiosis	Genetics.cpp:44	fn-not-found
v2-rust-core	Chiasma locations	.draw_meiosis	Genetics.cpp:53	fn-not-found

Group	Topic	Function	Cited location	Status
v2-rust-core	Chiasma locations	.draw_meiosis	Genetics.cpp:86–97	fn-not-found
v2-rust-core	Strand raffle	.draw_meiosis / chromosome_mask	Genetics.cpp:55–55	fn-not-found
v2-rust-core	Breakpoint rank	breaks_at	Genetics.cpp:67–67	fn-not-found
v2-rust-core	Gamete assembly	recombine	Genetics.cpp:356–356	fn-not-found
v2-rust-core	Cross	mate_haplotypes	cross_mating.R:102–102	fn-in-file-only
v2-rust-core	Genotype projection	write_genotype, dosages	cross_population.R:154–158	fn-in-file-only
v2-rust-core	Founder phasing	as_population	cross_population.R:118–119	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numericalize_core.rs:95	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numericalize_core.rs:96	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numericalize_core.rs:103	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numericalize_core.rs:110	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numericalize_core.rs:117	fn-in-file-only
v2-selection	Selection differential	select_ind	R/select_ind.R:281–281	fn-in-file-only
v2-selection	Realized intensity	select_ind, .select_culling	R/select_ind.R:282–283	fn-in-file-only
v2-selection	Independent culling	.select_culling	R/select_ind.R:712–726	fn-in-file-only
v2-selection	Index : culling : tandem	roxygen; test-culling.R:71–89	R/select_ind.R:97–104	fn-not-found
v2-selection/codex	Selfed / DH family BV correlation	select_ind() docs (family_relationship)	R/select_ind.R:69–73	fn-in-file-only
v2-selection/codex	Sequential culling	.select_culling	R/select_ind.R:712–719	fn-in-file-only
v2-selection/codex	Scheme RNG threading	all four wrappers, .self_each, .intermate	R/select_schemes.R:89–89	fn-in-file-only
v2-ocs-usefulness-marker	OCS objective	optimum_contribution, .frank_wolfe	R/select_ocs.R:100–103	fn-in-file-only
v2-ocs-usefulness-marker/codex	Ridge blend of G	g_matrix	R/select_ocs.R:25–26	fn-in-file-only

Group	Topic	Function	Cited location	Status
v2-prediction/codex	Known-variance mixed model (the model BLUP solves)	predict_ebv	R/select_blup.R:90	in-file-only
v2-transcriptome	Expression model	simulate_transcriptome	R/transcriptome_sim.R:490-492	in-file-only
v2-transcriptome	cis score	simulate_transcriptome	R/transcriptome_sim.R:381-391	in-file-only
v2-transcriptome	trans factor	simulate_transcriptome	R/transcriptome_sim.R:392-392	in-file-only
v2-transcriptome	Additive genetic score	simulate_transcriptome	R/transcriptome_sim.R:428-457	in-file-only
v2-transcriptome	Epistatic blend	simulate_transcriptome	R/transcriptome_sim.R:402-451	in-file-only
v2-transcriptome	Residual	simulate_transcriptome	R/transcriptome_sim.R:464-474	in-file-only
v2-transcriptome	Realized heritability	simulate_transcriptome, predict	R/transcriptome_sim.R:770-773	in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_sim.R:503-513	in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_sim.R:532-532	in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_sim.R:539-539	in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_sim.R:550-550	in-file-only
v2-transcriptome	Genetic budget	simulate_transcriptome	R/transcriptome_sim.R:518-525	in-file-only
v2-transcriptome	Realized cis fraction	simulate_transcriptome	R/transcriptome_sim.R:526-528	in-file-only
v2-transcriptome	Mimic affine	simulate_transcriptome	R/transcriptome_sim.R:481-490	in-file-only
v2-io-formats	Allele frequency / MAF	filter_geno	R/qc_filter_geno.R:203-210	in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:221-221	in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:224-224	in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:228-228	in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:211-211	in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:232-232	in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:234-234	in-file-only
v2-io-formats	Pruning decision	.ld_sweep	R/qc_filter_geno.R:498-498	in-file-only
v2-io-formats	Pruning decision	.ld_sweep	R/qc_filter_geno.R:500-503	in-file-only
v2-io-formats	Window / step	.ld_prune, .ld_sweep	R/qc_filter_geno.R:440-452	in-file-only
v2-io-formats	Window / step	.ld_prune, .ld_sweep	R/qc_filter_geno.R:617-617	in-file-only
v2-io-formats	Numeric coding	numericalize_core	src/rust/src/numericalize_core.rs:117	in-file-only

Group	Topic	Function	Cited location	Status
v1-core-linkage	Epistatic value	genetic_effect	legacy_genetic_effect.R:177-89	fn-in-file-only
v1-core-linkage	Centering	genetic_effect	legacy_genetic_effect.R:93	fn-in-file-only
v1-core-linkage	Degree of dominance	check_in	legacy_check_in.R:252	fn-in-file-only
v1-core-linkage	Geometric series	check_in	legacy_check_in.R:786	fn-in-file-only
v1-core-linkage	Residual variance	phenotypes	legacy_Phenotypes.R:216	fn-in-file-only
v1-core-linkage	Single-trait phenotype	phenotypes	legacy_Phenotypes.R:674	fn-in-file-only
v1-core-linkage	Multi-trait residual	phenotypes	legacy_Phenotypes.R:245	fn-in-file-only
v1-core-linkage	Null trait ($h^2=0$)	phenotypes	legacy_Phenotypes.R:534	fn-in-file-only
v1-core-linkage	Realized heritability	phenotypes	legacy_Phenotypes.R:251	fn-in-file-only
v1-core-linkage	Per-QTN PVE	phenotypes	legacy_Phenotypes.R:268	fn-in-file-only
v1-core-linkage	Residual seed	phenotypes	legacy_Phenotypes.R:234	fn-in-file-only
v1-core-linkage	LD-QTN seed	qtn_linkage	legacy_QTN_linkage.R:110	fn-in-file-only
v1-core-linkage	LD measure	qtn_linkage	legacy_QTN_linkage.R:151	fn-in-file-only
v1-core-linkage	Indirect selection	qtn_linkage	legacy_QTN_linkage.R:212	fn-in-file-only
v1-core-linkage	Direct selection	qtn_linkage	legacy_QTN_linkage.R:1308	fn-in-file-only
v1-core-linkage	Direct selection	qtn_linkage	legacy_QTN_linkage.R:1062	fn-in-file-only
v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	QTN_linkage.R:311-314	fn-in-file-only
v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	qtn_from_user.R:237-240	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:491	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:484	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:1336	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:488	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:1340	fn-in-file-only
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:113	fn-in-file-only
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:109	fn-in-file-only

Group	Topic	Function	Cited location	Status
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:113	in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1358-1371	in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1447-1460	in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1517-1530	in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1589-1602	in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1361-1361	in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1531-1554	in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1532-1532	in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1541-1541	in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1544-1544	in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1234-1237	in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:125-125	in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:976-983	in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1467-1473	in-file-only
v1-pleio-effects	Geometric effect series	check_in	R/legacy_check_in.R:77-837	in-file-only
v1-pleio-effects	Dominance genetic value	genetic_effect	R/legacy_check_in.R:252-252	found
v1-pleio-effects	Epistatic genetic value	genetic_effect	R/legacy_genetic_effect.R:70-91	in-file-only
v1-pleio-effects	Total genetic value	genetic_effect	R/legacy_genetic_effect.R:92-96	in-file-only
v1-pleio-effects	MAF of a QTN	qtn_pleiotropic, qtn_partially_pleiotropic, constraint	R/legacy_QTN_pleiotropic.R:161-165	in-file-only
v1-pleio-effects	QTN sampling seeds (pleiotropic)	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:130-130	in-file-only
v1-pleio-effects	Whitening	base_line_multi_traits	R/legacy_Base_line_multi_traits.R:103-109	in-file-only
v1-pleio-effects	Colouring & rescale	base_line_multi_traits	R/legacy_Base_line_multi_traits.R:110-120	in-file-only

Group	Topic	Function	Cited location	Status
v1-pleio-effects	Residual variance (context)	phenotypes	R/legacy_Phenotypes.R:216	in-file-only
v1-pleio-effects	vQTL sample heritability	vQTL	R/legacy_vQTL.R:184-193	in-file-only
v1-pleio-effects/codex	Fully pleiotropic QTN-set construction	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:125-157	in-file-only
v1-pleio-effects/codex	Partial-pleiotropy locus construction	qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:120-190	in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_pleiotropic.R:148-180	in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:616-641	in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:654-654	in-file-only

Appendix B – quick index (topic, function, location)

Ver	Group	Topic	Function	Location
V2	v2-grammar	Layer scaling to prop	.genetic_matrix	R/grammar_realize.R:139-161 (146)
V2	v2-grammar	Additive component	.component_raw	R/grammar_realize.R:498-509 (500), centred 432
V2	v2-grammar	Orthogonal genotypic value	.component_raw	R/grammar_realize.R:501-507 (506)
V2	v2-grammar	Dominance component	.component_raw	R/grammar_realize.R:510
V2	v2-grammar	Epistasis unit column	.epi_unit_column	R/grammar_realize.R:579-589 (586, 588)
V2	v2-grammar	Epistasis component	.component_raw	R/grammar_realize.R:511-529 (524)
V2	v2-grammar	Geometric effect series	.effect_series	R/effects_series.R:23-72 (64)
V2	v2-grammar	Repulsion phase	.apply_phase	R/grammar_layers.R:1482-1490 (1487)
V2	v2-grammar	Residual budget	.realize_phenotype	R/grammar_realize.R:55-59 (59)
V2	v2-grammar	Residual draw	.draw_residual	R/effects_series.R:85-102 (99)
V2	v2-grammar	Phenotype	.realize_phenotype	R/grammar_realize.R:96
V2	v2-grammar	Requested budget identity	.resolve_prop, .add_layer, .check_h2_complete	R/grammar_layers.R:1085-1098, 1102-1115; R/grammar_realize.R:1153-1166
V2	v2-grammar	One-call split	.build_one_call	R/grammar_simulate_phenotype.R:780 (769)
V2	v2-grammar	Realized H ²	.realized_h2	R/grammar_realize.R:1078-1106 (716-720)

Ver	Group	Topic	Function	Location
V2	v2-grammar	Average effect of substitution	.avg_effect	R/grammar_realize.R:182-184 (183)
V2	v2-grammar	Breeding value	.breeding_value_maf	R/grammar_realize.R:282-303 (227-235)
V2	v2-grammar	Orthogonal split	.orthogonal_var_sp .variance_budget	R/grammar_realize.R:778-794 (646-650), 536-551
V2	v2-grammar	HWE theory (check only)	— (audit check)	—
V2	v2-grammar	vQTL loading	.apply_vqtl	R/grammar_realize.R:605-618 (616)
V2	v2-grammar	vQTL residual	.apply_vqtl	R/grammar_realize.R:627-641
V2	v2-grammar	Transcriptome score	.tx_raw, .transcriptome_matrix	R/grammar_realize.R:454-476 (470, 473, 475), 308-338
V2	v2-grammar	Mediation split	.mediation_budget	R/grammar_realize.R:738-750 (604-606)
V2	v2-grammar	Complex combination	complex_phenotypes	R/grammar_complex.R:122-134 (123, 127), 96-107 (99)
V2	v2-grammar	MAF	.marker_maf_ref	R/grammar_simulate_phenotype 1052 (1040, 1044)
V2	v2-grammar	Sub-seed	.layer_seed	R/grammar_simulate_phenotype 1094 (1083, 1092)
V2	v2-grammar	Per-QTN variance	.qtn_var	R/io_write.R:1288-1312 (217-218)
V2	v2-grammar	Fixed-scale phenotype	phenotype_value	R/cross_population.R:1004-1104 (1061)
V2	v2-grammar	Genetic correlation (plot)	.plot_cor	R/grammar_plot.R:144-162 (150)
V2	v2-grammar (codex)	PleioArch shared covariance	.pleio_draw, .pleio_nonadditive	R/effects_pleioarch.R:65-93 (sigma at 78-80), 345-358 (356-357)
V2	v2-grammar (codex)	PleioArch effect allocation (non-additive)	.pleio_unit_effects, .draw_mvnorm	R/effects_pleioarch.R:583-690 (535-537), 780-800
V2	v2-grammar (codex)	PleioArch attainability	.check_pleio_feasible	R/effects_pleioarch.R:870-917
V2	v2-grammar (codex)	Total pleiotropic correlation target	.pleio_total_cor_check	R/effects_pleioarch.R:724-788 (roxygen 543-574)
V2	v2-grammar (codex)	Additive MAF effect scaling	.pleio_draw	R/effects_pleioarch.R:201-217 (144-149)
V2	v2-grammar (codex)	LD window (two-trait linked distinct loci)	.draw_qtn_ld	R/arch_ld.R:62-225 (window 69-95, sampling 105-175)
V2	v2-effects-arch	Pleiotropic covariance	.pleio_draw, .pleio_nonadditive	effects_pleioarch.R:90-91 draw 504-505
V2	v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	:197-199, :685
V2	v2-effects-arch	Major/minor split	.pleio_draw, .draw_mvnorm	:195-196, :934
V2	v2-effects-arch	MVN draw	.draw_mvnorm	:929-939
V2	v2-effects-arch	Univariate draw	.draw_univariate	:944-949

Ver	Group	Topic	Function	Location
V2	v2-effects-arch	Shared-unit count	.pleio_partition	:291–292, 320
V2	v2-effects-arch	Attainability (2 traits)	.check_pleio_feasib	876–890
V2	v2-effects-arch	Attainability (n traits)	.check_pleio_feasib	898–915
V2	v2-effects-arch	Allele -> genotype scaling	.pleio_draw; .marker_maf_ref	:201–217; grammar_simulate_phenotype 1044
V2	v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	601–604, 686; grammar_realize.R:579– 589; dominance design :518 vs gram- mar_realize.R:510
V2	v2-effects-arch	Effective per-component target	.pleio_unit_effects	659–664
V2	v2-effects-arch	Total-correlation target	.pleio_total_cor_ch	754–759
V2	v2-effects-arch	Layer rescale (context)	.genetic_matrix	grammar_realize.R:144– 146
V2	v2-effects-arch	Few-unit attenuation	doc claim	:26–28; grammar_simulate_phenotype 320
V2	v2-effects-arch	Complete-LD ensemble mean	doc claim	:28–30; DECISIONS.md 023 Scope
V2	v2-effects-arch	Geometric effect series	.effect_series	effects_series.R:55, 64
V2	v2-effects-arch	Repulsion phase	.apply_phase	grammar_layers.R:1482– 1490
V2	v2-effects-arch	Residual draw	.draw_residual	effects_series.R:85– 102
V2	v2-effects-arch	LD measure	window_partners, r2_pair	arch_ld.R:111, 124–126
V2	v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:149– 160 (:151–152); partner :73–74
V2	v2-effects-arch	Indirect flanks	.draw_qtn_ld	arch_ld.R:161– 197
V2	v2-effects-arch	Layer sub-seed	.layer_seed	grammar_simulate_phenotype 1093
V2	v2-effects-arch	Gabriel blocks tag	.gabriel_blocks	qc_ld_methods.R:24– 29, 38–43
V2	v2-effects-arch (codex)	Zero-variance correlation guard	.pleio_check_zero_var_eff	R/effects_pleioarch.R:246– 275
V2	v2-effects-arch (codex)	Shared/specific unit count (clamped)	.pleio_partition()	R/effects_pleioarch.R:288– 329 (clamp :291–292)
V2	v2-effects-arch (codex)	One shared unit	.pleio_single_unit	R/effects_pleioarch.R:412– 441
V2	v2-effects-arch (codex)	Correlation input expansion	.pleio_cor_matrix()	R/effects_pleioarch.R:957– 986
V2	v2-effects-arch (codex)	Pleiotropic-share argument mapping	.pleio_pi_vector()	R/effects_pleioarch.R:829– 860

Ver	Group	Topic	Function	Location
V2	v2-effects-arch (codex)	Architecture-specific QTN sampling	.draw_qtn()	R/arch_independent.R:21–45
V2	v2-effects-arch (codex)	Distinct-chromosome allocation	.draw_qtn_distinct	R/arch_independent.R:56–75
V2	v2-effects-arch (codex)	Epistatic-set sampling	.draw_qtn_pairs()	R/arch_independent.R:81–103
V2	v2-effects-arch (codex)	Candidate-marker rule	.candidate_markers	R/arch_independent.R:114–134
V2	v2-effects-arch (codex)	Layer occurrence index	.type_occurrence()	R/arch_independent.R:139–141
V2	v2-crossing	Heterozygote phasing	as_population	R/cross_population.R:176–177
V2	v2-crossing	Dosage from strands	dosages	R/cross_population.R:623; genome.rs:273–278
V2	v2-crossing	Crossover count	.draw_meiosis	R/cross_mating.R:53–56 (dispatch), Poisson branch :77–78; gamma model :215–275; default .INTERFERENCE_DEFAULT :117 (DECISION-047); cM→M :339,341
V2	v2-crossing	Chiasma positions	.draw_meiosis	R/cross_mating.R:79–87
V2	v2-crossing	Strand choice	.draw_meiosis	R/cross_mating.R:90
V2	v2-crossing	Ancestry mask	chromosome_mask, breaks_at	src/rust/src/meiosis.rs:4951, 63–72
V2	v2-crossing	Gamete	recombine	meiosis.rs:224–237
V2	v2-crossing	Cross / self / DH	mate_haplotypes	meiosis.rs:350–371; events per progeny R/cross_mating.R:345
V2	v2-crossing	Recombination fraction (implied, verified)	— (property of the Poisson process)	verified e4.R/e5.R; tests/testthat/test-cross.R:179–202 (file runs under simplePHENOTYPES.interference = "poisson", :13)
V2	v2-crossing	Heterozygosity under selfing	selfcross (doc)	R/cross_mating.R:627–629; verified e1.R E3
V2	v2-crossing	Synthetic map rate	synthetic_map	R/cross_map.R:157
V2	v2-crossing	Synthetic map integration	synthetic_map	R/cross_map.R:168–173, :133
V2	v2-crossing	Additive value (fixed scale)	additive_value	R/cross_population.R:746
V2	v2-crossing	Genotypic value	genotypic_value	R/cross_population.R:821
V2	v2-crossing	Residual from h^2	phenotype_value	R/cross_population.R:1061

Ver	Group	Topic	Function	Location
V2	v2-crossing	Breed composition	breed_composition	R/cross_breed.R:37-43
V2	v2-crossing	Realized heterosis	heterosis	R/cross_breed.R:135-139, 152
V2	v2-crossing	Expected F1 heterosis	heterosis + .expected_cross_means	R/cross_breed.R:140-150; R/select_combining.R:244-255
V2	v2-crossing	Per-locus cross mean	.expected_cross_means	R/select_combining.R:249-254
V2	v2-crossing	Retention fractions (doc only)	roxygen	R/cross_breed.R:68-72; docs/SPEC-block3b.md §7
V2	v2-crossing	Rotation sire sequence	crossbreed	R/cross_breed.R:302-305
V2	v2-crossing	Pedigree key	.mating_pedigree, .stable_key	R/cross_pedigree.R:25-37 (.stable_key), :42-59 (.key_part), :205-207 (mating key); hash.rs:12-23
V2	v2-crossing	Generation	.mating_pedigree	R/cross_pedigree.R:208-209
V2	v2-crossing	A-matrix (other group, consistency only)	a_matrix	R/select_blup.R:69-90
V2	v2-crossing (codex)	Average-effect vs genotypic-value distinction (doc)	roxygen of genotypic_value (implemented in select_ind(on = "bv"), other group)	R/cross_population.R:774-780 (verified: a text at 776-777)
V2	v2-crossing (codex)	Full-sib family identity	families	R/cross_pedigree.R:307-336 (verified: families <- at 307, closes at 336)
V2	v2-crossing (codex)	Mating designs (counts)	mating_design	R/cross_mate.R:360-486 (verified: mating_design <- at 229; file is 350 lines)
V2	v2-crossing (codex)	Within-family relationship validation (test)	test helper .icc	tests/testthat/test-family-relationship.R:1-58 (verified: 58 lines, .icc at 26, targets at 52-57)
V2	v2-rust-core	Crossover count (count-location)	.draw_meiosis	R R/cross_mating.R:77-78; isqg Genet-ics.cpp:44, 84; Rust: none (input counts)

Ver	Group	Topic	Function	Location
V2	v2-rust-core	Chiasma locations	.draw_meiosis	R : 79–87; isqg Genetics.cpp:86–97
V2	v2-rust-core	Strand raffle	.draw_meiosis / chromosome_mask	R : 90; Rust meiosis.rs:68–70; isqg : 74–75
V2	v2-rust-core	Breakpoint rank	breaks_at	Rust meiosis.rs:49–51; isqg Genetics.cpp:67
V2	v2-rust-core	Ancestry mask	chromosome_mask, Bits::toggle_from, flip_all	Rust meiosis.rs:63–72, genome.rs:69–87; isqg : 58–70 (XOR loop), : 74–75 (flip)
V2	v2-rust-core	Gamete assembly	recombine	Rust meiosis.rs:224–237; isqg Genetics.cpp:356
V2	v2-rust-core	Cross	mate_haplotypes	Rust meiosis.rs:359–369; R .mate cross_mating.R:345,453,460–467
V2	v2-rust-core	Self	selfcross → .mate(parent, parent)	R cross_mating.R:647–652; Rust same path
V2	v2-rust-core	Doubled haploid	mate_haplotypes (Dh)	Rust meiosis.rs:263–269,351–358; R cross_mating.R:686–691
V2	v2-rust-core	Genotype projection	write_genotype, dosages	Rust genome.rs:273–284; R cross_population.R:623 (cis + trans - 1)
V2	v2-rust-core	Founder phasing	as_population	R cross_population.R:176–177
V2	v2-rust-core	Map units	.mate	R cross_mating.R:339,341
V2	v2-rust-core	Haldane map function (consequence, tested only)	—	test tests/testthat/test-cross.R:179–202; auditor T2–T4
V2	v2-rust-core	Numericalization — orientation	compute_flip	R io_detect_format.R:362–364; reference mode : 354–359
V2	v2-rust-core	Numericalization — codes	numericalize_core	Rust numeric.rs:76–88,135–180

Ver	Group	Topic	Function	Location
V2	v2-rust-core	Pedigree key hash	fnv1a_128, .stable_key	Rust hash.rs:12–23; R cross_pedigree.R:22– 59
V2	v2-rust-core (codex)	Public ingestion dispatch	as_numeric()	R/io_as_numeric.R:192– 207 (verified: if (is.character(x) && is.null(dim(x))) at:198, for- mat_conversion(...) at:206)
V2	v2-rust-core (codex)	Genetic-model post-transform (split out of Fable’s single “codes” row)	numericalize_core()	src/rust/src/numeric.rs:14 180 (verified: match model at:156, "Dom":157, "Left" :164, "Right":171, default:178)
V2	v2-selection	Truncation count from prop	.resolve_keep	R/select_ind.R:574
V2	v2-selection	Selection intensity (large N)	.resolve_keep	R/select_ind.R:585– 587
V2	v2-selection	Selection differential	select_ind	R/select_ind.R:460, 482
V2	v2-selection	Realized intensity	select_ind, .select_culling	R/select_ind.R:461– 478, 1089–1090
V2	v2-selection	Response (documented, tested here)	roxygen only	R/select_ind.R:16– 21
V2	v2-selection	Average effect	.avg_effect	R/grammar_realize.R:182– 184
V2	v2-selection	Breeding value	.breeding_value_max	R/grammar_realize.R:236– 305 (loop 282–303)
V2	v2-selection	Smith–Hazel index	.index_score, .index_weights	R/select_ind.R:769– 782; 799–842
V2	v2-selection	QGSi	.quadratic_index_score	R/select_ind.R:737, 744–747
V2	v2-selection	Lush combined index	.combined_score	R/select_ind.R:867, 872–877, 889–902
V2	v2-selection	Within-family allocation	.sel_within_family	R/select_ind.R:939– 949
V2	v2-selection	Among-family	.sel_among_family	R/select_ind.R:968– 978
V2	v2-selection	Independent culling	.select_culling	R/select_ind.R:1067– 1081
V2	v2-selection	Index : culling : tandem	roxygen; test-culling.R:71–89	R/select_ind.R:157– 164
V2	v2-selection	Bulk pool	bulk	R/select_schemes.R:156– 169; .bulk_counts :470–472
V2	v2-selection	Pedigree family size	pedigree	R/select_schemes.R:304– 307
V2	v2-selection	Tandem schedule	pedigree, recur- rent_selection	R/select_schemes.R:291, 398

Ver	Group	Topic	Function	Location
V2	v2-selection	Intermating	.intermate	R/select_schemes.R:542–545
V2	v2-selection (codex)	General index response (context, not implemented)	roxygen only	R/select_ind.R:16–21
V2	v2-selection (codex)	Singular Smith–Hazel extension	.index_weights	R/select_ind.R:799–830
V2	v2-selection (codex)	Selfed / DH family BV correlation	select_ind() docs (family_relationship)	R/select_ind.R:94–98
V2	v2-selection (codex)	Sequential culling	.select_culling	R/select_ind.R:1067–1074
V2	v2-selection (codex)	Population pooling	c.Population	R/select_schemes.R:25–68
V2	v2-selection (codex)	Single seed descent	single_seed_descent	R/select_schemes.R:102–118
V2	v2-selection (codex)	Recurrent-selection cycle	recurrent_selection .intermate	R/select_schemes.R:387–410; 354–370
V2	v2-selection (codex)	Scheme RNG threading	all four wrappers, .self_each, .intermate	R/select_schemes.R:111, 155, 283, 384; 521; 544
V2	v2-ocs-usefulness-marker	Genomic relationship	g_matrix	R/select_ocs.R:71–92
V2	v2-ocs-usefulness-marker	Genomic inbreeding	g_matrix (doc)	R/select_ocs.R:16, 42–43
V2	v2-ocs-usefulness-marker	OCS objective	optimum_contribution .frank_wolfe	R/select_ocs.R:105–108, 601–624
V2	v2-ocs-usefulness-marker	Group coancestry	optimum_contribution .tune_lambda	R/select_ocs.R:319, 666
V2	v2-ocs-usefulness-marker	FW gradient / duality gap	.frank_wolfe	R/select_ocs.R:601–607
V2	v2-ocs-usefulness-marker	Away-step and line search	.frank_wolfe	R/select_ocs.R:609–625
V2	v2-ocs-usefulness-marker	Penalty tuning	.tune_lambda	R/select_ocs.R:652–718
V2	v2-ocs-usefulness-marker	Parent sampling	sample_parents	R/select_ocs.R:429–441; .allocate_slots :473–494
V2	v2-ocs-usefulness-marker	Usefulness	cross_usefulness	R/select_usefulness.R:142–144
V2	v2-ocs-usefulness-marker	Selection intensity	.intensity_from_p	R/select_usefulness.R:161–163
V2	v2-ocs-usefulness-marker	Fixed additive score	.additive_model, .additive_gv	R/select_usefulness.R:204–209, 228
V2	v2-ocs-usefulness-marker	Average effect	.avg_effect	R/grammar_realize.R:182–184
V2	v2-ocs-usefulness-marker	MAS feasibility	marker_select	R/select_marker.R:116–119
V2	v2-ocs-usefulness-marker	MARS/oracle index	additive_value (documented in marker_select)	R/cross_population.R:735–747; doc R/select_marker.R:20–33

Ver	Group	Topic	Function	Location
V2	v2-ocs-usefulness-marker	Recurrent-allele score	.mabc_founders	R/select_mabc.R:311
V2	v2-ocs-usefulness-marker	Recovery	.mabc_recovery	R/select_mabc.R:464
V2	v2-ocs-usefulness-marker	Expected recovery	doc only	R/select_mabc.R:25–31, 227–229
V2	v2-ocs-usefulness-marker	Interval weights	.mabc_weights	R/select_mabc.R:520–529
V2	v2-ocs-usefulness-marker	Foreground / recombinant / background staging	mabc_select	R/select_mabc.R:170–177, 194
V2	v2-ocs-usefulness-marker (codex)	Ridge blend of G	g_matrix	R/select_ocs.R:25–26, 44–50, 93–95
V2	v2-ocs-usefulness-marker (codex)	MABC foreground feasibility	mabc_select	R/select_mabc.R:168–174
V2	v2-prediction	Tabular relationship (off-diagonal)	a_matrix	R/select_blup.R:73–79
V2	v2-prediction	Inbreeding / diagonal	a_matrix	R/select_blup.R:80–88
V2	v2-prediction	Variance ratio	predict_ebv, .blup_variances	R/select_blup.R:365, :676–681
V2	v2-prediction	GLS mean	predict_ebv	R/select_blup.R:376–386
V2	v2-prediction	BLUP of u	predict_ebv	R/select_blup.R:387–388
V2	v2-prediction	PEV / reliability	predict_ebv	R/select_blup.R:395–398
V2	v2-prediction	GBLUP marker back-solve	.gblup_marker_effects	R/select_blup.R:695–703
V2	v2-prediction	Accuracy / bias	prediction_accuracy	R/select_blup.R:752–761
V2	v2-prediction	Expected cross mean (one locus)	.expected_cross_mean	R/select_combining.R:244–255
V2	v2-prediction	Tester-referenced average effect	combining_ability (doc)	R/select_combining.R:30–37
V2	v2-prediction	Topcross / factorial GCA, SCA	.decompose_ca	R/select_combining.R:321–324
V2	v2-prediction	Diallel GCA (method 4)	.decompose_ca	R/select_combining.R:312–316
V2	v2-prediction	Diallel SCA (method 4)	.decompose_ca	R/select_combining.R:317–318
V2	v2-prediction	Broad-sense residual for simulated crosses	phenotype_value via .simulate_cross_means	R/cross_population.R:1061; R/select_combining.R:290–293
V2	v2-prediction	Realized-scale template effects	.layer_scaled_effects ← template_effects	R/grammar_realize.R:364; R/select_combining.R:400–406
V2	v2-prediction	Progeny-test expected mean	progeny_test (doc)	R/select_progeny.R:17–35
V2	v2-prediction	Progeny-test accuracy	progeny_test (doc)	R/select_progeny.R:37–53
V2	v2-prediction	Mate sampling	progeny_test	R/select_progeny.R:121–135
V2	v2-prediction	Family mean	progeny_test	R/select_progeny.R:149–150

Ver	Group	Topic	Function	Location
V2	v2-prediction (codex)	Known-variance mixed model (the model BLUP solves)	predict_ebv	R/select_blup.R:174- 181 (roxygen), :300-321 (inputs)
V2	v2-prediction (codex)	Supplied- covariance validation (correlation-scale PSD test)	.check_relationship	R/select_blup.R:587- 650 (scaling :630-632, eigen :640, tolerance :644-645)
V2	v2-prediction (codex)	VanRaden method-1 genomic relationship (consumed by GBLUP)	predict_ebv → g_matrix	R/select_blup.R:337- 343 (Codex wrote 168-173); R/select_ocs.R:44- 98 (Codex wrote 44-92; core at :83, 90-92)
V2	v2-prediction (codex)	Progeny-test RNG draw order (reproducibility contract)	progeny_test	R/select_progeny.R:113- 114 (seed), :132 (mates), :138 (meioses), :144-145 (residual)
V2	v2-transcriptome	Expression model	simulate_transcriptome	R/transcriptome_simulate.R:618
V2	v2-transcriptome	Reference-centered dosage	simulate_transcriptome	:308-310; predict :845-847
V2	v2-transcriptome	cis score	simulate_transcriptome	:504-511
V2	v2-transcriptome	trans factor	simulate_transcriptome	:454-471, :512
V2	v2-transcriptome	Standardization	z1	:492-496
V2	v2-transcriptome	Additive genetic score	simulate_transcriptome	:548-578
V2	v2-transcriptome	Epistatic blend	simulate_transcriptome	:522-572
V2	v2-transcriptome	Residual	simulate_transcriptome	:585-595
V2	v2-transcriptome	Realized heritability	simulate_transcriptome predict	:610-628; :943-948
V2	v2-transcriptome	Effective coefficients	simulate_transcriptome	:636-646, :666, :673, :684
V2	v2-transcriptome	Genetic budget	simulate_transcriptome	:654-658
V2	v2-transcriptome	Realized cis fraction	simulate_transcriptome	:659-661; predict :950
V2	v2-transcriptome	Cross-population genetic value	predict.transcriptome	:829-848, :860-886
V2	v2-transcriptome	Mimic affine	simulate_transcriptome	:602-616
V2	v2-transcriptome	GRM	.tx_grm	R/transcriptome_mimic.R:24 30
V2	v2-transcriptome	REML objective	.greml_h2	R/transcriptome_mimic.R:76 84
V2	v2-transcriptome	GREML heritability	.greml_h2	R/transcriptome_mimic.R:94 97
V2	v2-transcriptome	Factor count	.tx_estimate_factor	R/transcriptome_mimic.R:11 119
V2	v2-transcriptome	kappa proxy	.tx_estimate_kappa	R/transcriptome_mimic.R:14 172 (roxygen :120-148); called at R/transcriptome_simulate.R
V2	v2-transcriptome	NB observation	observe_counts	R/transcriptome_counts.R:7 80, :98-102

Ver	Group	Topic	Function	Location
V2	v2-transcriptome	Layer score	.tx_raw, .transcriptome_matrix	R/grammar_realize.R:454-476, :414-440
V2	v2-transcriptome	Mediation split	.tx_raw(which="genotype"), .mediation_budget, mediation_split	R/grammar_realize.R:458-476, :730-761; R/io_write.R:1267-1270
V2	v2-transcriptome	Realized H2	.genetic_value_matrix, .realized_h2	R/grammar_realize.R:394-400, :1078-1106
V2	v2-transcriptome	Per-gene share	.tx_qtn_var	R/io_write.R:1328-1345
V2	v2-transcriptome	Layer sub-seed	.layer_seed, transcriptome	R/grammar_simulate_phenotype.R:1092; R/transcriptome_layer.R:252-276
V2	v2-transcriptome (codex)	Single-component LMM (model statement)	.grem1_h2	R/transcriptome_mimic.R:3253 (roxygen), fit :58-97
V2	v2-transcriptome (codex)	Calibration metrics	benchmark 01	benchmarks/01_h2_calibration (h2), :93-94 (cis)
V2	v2-transcriptome (codex)	Marginal eQTL scan	scan_one	benchmarks/02_eqtl_recovery (r at :75, best rank :80)
V2	v2-transcriptome (codex)	Co-expression rank test	benchmark 03	benchmarks/03_coexpression_rank :70
V2	v2-transcriptome (codex)	Benchmark-04 mediation target	benchmark 04	benchmarks/04_mediation_rank :12, 89-90, 114-116
V2	v2-transcriptome (codex)	TWAS Pearson test	cor_p	benchmarks/05_twas_power.R:112 (t at :110, p at :111)
V2	v2-transcriptome (codex)	Bonferroni threshold	benchmark 05	benchmarks/05_twas_power.R:146-147
V2	v2-io-formats	Allele frequency / MAF	filter_geno	R/qc_filter_geno.R:290-297
V2	v2-io-formats	Monomorphic / MAF filters	filter_geno	:311, :314, :318
V2	v2-io-formats	Heterozygosity	filter_geno	:301, :322, :324
V2	v2-io-formats	Composite (genotypic) r^2	.ld_sweep	:556-565
V2	v2-io-formats	Pruning decision	.ld_sweep	:532, :663, :665-668
V2	v2-io-formats	Window / step	.ld_prune, .ld_sweep	:492-506, :605-617 (win_at), :737-782 (slide)
V2	v2-io-formats	Haplotypic r^2	.plink_hap_rsqs, .plink_em_hethet, .plink_cubic_roots	:796-809 (r^2 at :808), :822-879, :902-943
V2	v2-io-formats	VIF	.ld_sweep (vif branch)	:683-736 (solve :697, :721; test :725)
V2	v2-io-formats	Gabriel block MAF floor / tag	.gabriel_blocks	R/qc_ld_methods.R:25, :40
V2	v2-io-formats	Major-allele flip (frequency)	compute_flip	R/io_detect_format.R:362-364
V2	v2-io-formats	Reference flip	compute_flip	:359

Ver	Group	Topic	Function	Location
V2	v2-io-formats	Numeric coding	numericalize_core	src/rust/src/numeric.rs:76, :138–145, :156–178
V2	v2-io-formats	HWE	—	—
V2	v2-io-formats	Missing-rate filter	—	—
V2	v2-io-formats (codex)	Raw genotype parsing and coding dispatch	parse_hapmap_chars .apply_coding()	R/ionanet_format.R:223–326; R/io_read_formats.R:18–194
V2	v2-io-formats (codex)	Exact two-locus ML cubic	.plink_em_hethet()	R/qc_filter_geno.R:822–879
V2	v2-io-formats (codex)	Two-locus haplotype log-likelihood	.plink_calc_lnlike	R/qc_filter_geno.R:884–894
V2	v2-io-formats (codex)	Cubic real roots	.plink_cubic_roots	R/qc_filter_geno.R:902–943
V2	v2-io-formats (codex)	D' likelihood surface and class codes	.plink_calc_lnlike_quantile(), .plink_blocks_class	R/qc_filter_geno.R:952–966; :978–1053
V2	v2-io-formats (codex)	Gabriel/Haploview block acceptance	.plink_blocks_chrom .gabriel_blocks()	R/qc_filter_geno.R:1066–1180; R/qc_ld_methods.R:23–46
V2	v2-io-formats (codex)	Per-QTN marginal realized variance share	.qtn_var(), qtn_table()	R/io_write.R:1288–1312; :1400–1403
V2	v2-io-formats (codex)	Per-gene transcriptome marginal share	.tx_qtn_var()	R/io_write.R:1328–1345
V2	v2-io-formats (codex)	PLINK fixture allele serialization	fixture generator	data-raw/plink_parity_fixtures.34
V1	v1-core-linkage	Additive genetic value	genetic_effect	legacy_genetic_effect.R:10112
V1	v1-core-linkage	Dominance value	genetic_effect	:61–75
V1	v1-core-linkage	Epistatic value	genetic_effect	:77–89
V1	v1-core-linkage	Centering	genetic_effect	:93
V1	v1-core-linkage	Degree of dominance	check_in	legacy_check_in.R:274
V1	v1-core-linkage	Geometric series	check_in	:786–788, 796–798, 810–812, 823–825
V1	v1-core-linkage	Residual variance	phenotypes	legacy_Phenotypes.R:216, 650, 1065, 1511
V1	v1-core-linkage	Single-trait phenotype	phenotypes	:669–674, 1517–1522
V1	v1-core-linkage	Multi-trait residual	phenotypes	:238–245, 1080–1086
V1	v1-core-linkage	Null trait ($h^2=0$)	phenotypes	:533–534, 86–92
V1	v1-core-linkage	Realized heritability	phenotypes	:248–251, 336, 675–676, 876
V1	v1-core-linkage	Per-QTN PVE	phenotypes	:260–268, 685–693
V1	v1-core-linkage	Residual seed	phenotypes	:234, 665, 1076, 1513; 82, 529, 924, 1374

Ver	Group	Topic	Function	Location
V1	v1-core-linkage	LD-QTN seed	qtn_linkage	legacy_QTN_linkage.R:110, 383, 645, 971, 1220, 1243, 1455, 1479
V1	v1-core-linkage	LD measure	qtn_linkage	:150–151, 176–177, 1025–1028, 1270–1273
V1	v1-core-linkage	Indirect selection	qtn_linkage	:125–212
V1	v1-core-linkage	Direct selection	qtn_linkage	:1229–1308 (fixed), :978–1062, 1465–1546 (unfixed)
V1	v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	legacy_constraint.R:47–50; QTN_linkage.R:314–317; qtn_from_user.R:237–240
V1	v1-core-linkage	Genetic correlation (given cor)	base_line_multi_trait	legacy_Base_line_multi_trait
V1	v1-core-linkage (codex)	Heterozygote eligibility indicator	(not in group) constraint() (correct); create_phenotypes() guard (wrong code) phenotypes()	183 legacy_constraint.R:28–42; legacy_create_phenotypes.R:102, 848–935
V1	v1-core-linkage (codex)	Reported residual “sample correlation”		legacy_Phenotypes.R:483–491, 1335–1343 (verified: matrix(0,...) at :484/:1336, cov2cor at :488/:1340)
V1	v1-core-linkage (codex)	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:109–113, 382–386, 644–648, 970–974, 1219–1223, 1454–1458 (verified: while (s <= 10 & border) at :109,382,644,970,1219,1454 sample(index,...) at :113,386,648,974,1223,1458)
V1	v1-core-linkage (codex)	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:135–1388 (add), :1447–1460 (dom), :1517–1530 (epi), :1589–1602 (var) (verified: selected_snps <- genotypes\$snp %in% i at :1361, 1450, 1520, 1592)

Ver	Group	Topic	Function	Location
V1	v1-core-linkage (codex)	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:158– 1605 (verified: e_len <- lengths(QTN_list\$epi)/epi_ at :1532, each = epi_interaction at :1541, each = e_len[i] at :1544)
V1	v1-core-linkage (codex)	<i>(reconciler-added)</i> Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1269– 1269 (fixed add branch); same pattern :123–125 (indirect add), :976–983, :1467–1473
V1	v1-pleio-effects	Additive genetic value	genetic_effect	R/legacy_genetic_effect.R:57
V1	v1-pleio-effects	Geometric effect series	check_in	R/legacy_check_in.R:905– 837
V1	v1-pleio-effects	Dominance genetic value	genetic_effect	R/legacy_genetic_effect.R:77; R/legacy_check_in.R:274
V1	v1-pleio-effects	Epistatic genetic value	genetic_effect	R/legacy_genetic_effect.R:91
V1	v1-pleio-effects	Total genetic value	genetic_effect	R/legacy_genetic_effect.R:96
V1	v1-pleio-effects	Component variances	genetic_effect	R/legacy_genetic_effect.R:
V1	v1-pleio-effects	MAF of a QTN	qtn_pleiotropic, qtn_partially_pleiotropic, constraint	R/legacy_QTN_pleiotropic.R:165– 169; ...partially....R:165– 169; R/legacy_constraint.R:47– 43
V1	v1-pleio-effects	QTN sampling seeds (pleiotropic)	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:
V1	v1-pleio-effects	QTN sampling seeds (partially)	qtn_partially_pleiotropic	R/legacy_QTN_pleiotropic.R:125,153,446, ...partially....R:125,153,446,
V1	v1-pleio-effects	Het re-sample loop	both QTN samplers	R/legacy_QTN_pleiotropic.R:145,324–333; ...partially....R:131– 140,160–170,452– 461,481–493
V1	v1-pleio-effects	Whitening	base_line_multi_traits	R/legacy_Base_line_multi_t
V1	v1-pleio-effects	Colouring & rescale	base_line_multi_traits	R/legacy_Base_line_multi_t
V1	v1-pleio-effects	PD repair	make_pd	R/legacy_make_pd.R:61– 21
V1	v1-pleio-effects	Residual variance (context)	phenotypes	R/legacy_Phenotypes.R:216,
V1	v1-pleio-effects	vQTL residual scale	vQTL	R/legacy_vQTL.R:89,62– 71,76–89
V1	v1-pleio-effects	vQTL sample heritability	vQTL	R/legacy_vQTL.R:251– 193

Ver	Group	Topic	Function	Location
V1	v1-pleio-effects	Numeric normalisation	genotypes/numeric_07	R/legacy_Genotypes.R:79–62
V1	v1-pleio-effects	maf_cutoff filter	genotypes	R/legacy_Genotypes.R:163–111
V1	v1-pleio-effects (codex)	Fully pleiotropic QTN-set construction	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:201 (add), 313–345 (dom), 403–423 (var), 480–500 (epi)
V1	v1-pleio-effects (codex)	Partial-pleiotropy locus construction	qtn_partially_pleio	R/legacy_QTN_partially_pleio.R:233 (add+dom-loop), 294–360 (add only), 441–533 (dom), 616–684 (epi)
V1	v1-pleio-effects (codex)	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleio	R/legacy_QTN_pleiotropic.R:582opic R/legacy_QTN_partially_pleio.R:611 (also :654)

Appendix C — line remapping from c511c6f to e666a2c

Every reference in the entries and in Appendix B was moved to e666a2c with `git diff -U0 -M c511c6f` line mapping (Appendix A is kept as it was at the audit commit). 35 references were unchanged, 396 shifted with unchanged content, and 101 point into lines that were edited after the audit (listed below: the equation there may have changed and must be re-checked against the code before the port relies on it). 33 references name a file outside this repository or an ambiguous shorthand and were left as written.

Audit reference	Current reference
R/grammar_realize.R:447–455	R/grammar_realize.R:578–588
R/grammar_realize.R:409–425 (422)	R/grammar_realize.R:510–528 (523)
R/effects_series.R:20–53 (52)	R/effects_series.R:23–72 (64)
R/grammar_realize.R:36–40 (40)	R/grammar_realize.R:54–58 (58)
R/grammar_realize.R:704–723	R/grammar_realize.R:1064–1092
R/grammar_simulate_phenotype.R:649–670	R/grammar_simulate_phenotype.R:1000–1030
R/effects_pleioarch.R:53–81	R/effects_pleioarch.R:65–93
R/effects_pleioarch.R:143–159	R/effects_pleioarch.R:201–217
grammar_simulate_phenotype.R:666–669	grammar_simulate_phenotype.R:1018–1022
grammar_realize.R:447–455	grammar_realize.R:578–588
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